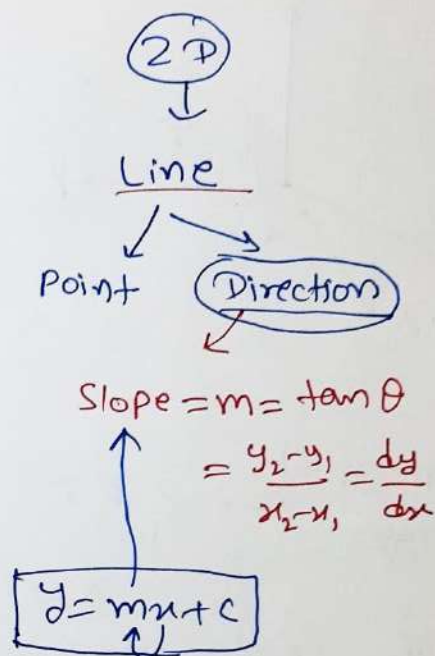
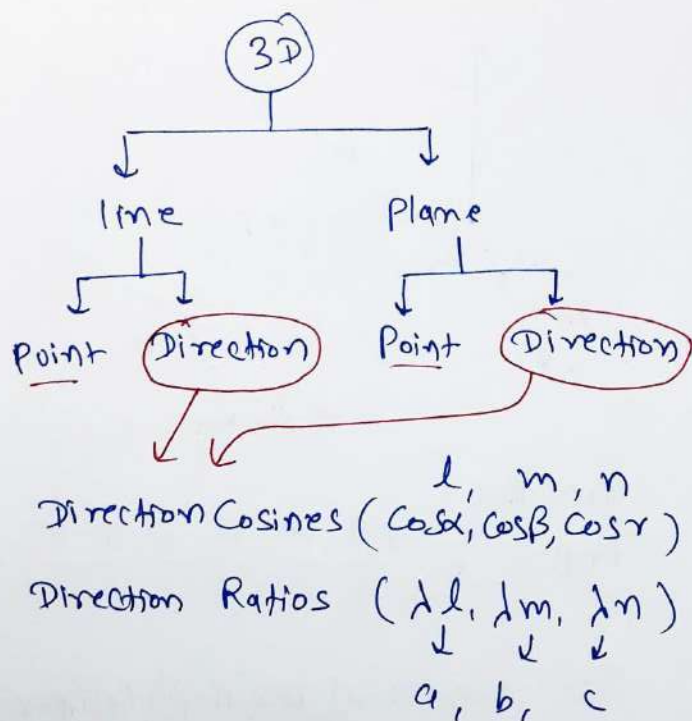


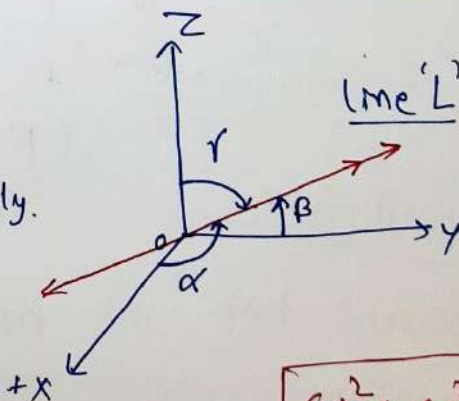
Direction Cosines (D.C.) & Direction Ratios (D.R.)

(दिक्कोण और दिक्अनुपात)

Direction Cosines (D.C.) & Direction Ratios (D.R.)

$\alpha, \beta, \gamma \rightarrow$  Direction Angles  
between Line 'L'  
and  $OX, OY, OZ$  respectively.

$(+x) \quad (+y) \quad (+z)$



$\left( \begin{matrix} \cos \alpha, \cos \beta, \cos \gamma \\ \downarrow \quad \downarrow \quad \downarrow \\ l \quad m \quad n \end{matrix} \right) \rightarrow$  Direction Cosines

$\left( \begin{matrix} \lambda \cos \alpha, \lambda \cos \beta, \lambda \cos \gamma \\ \lambda l, \lambda m, \lambda n \\ a, b, c \end{matrix} \right) \rightarrow$  Direction Ratios  
(Proportional to the D.C.)

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

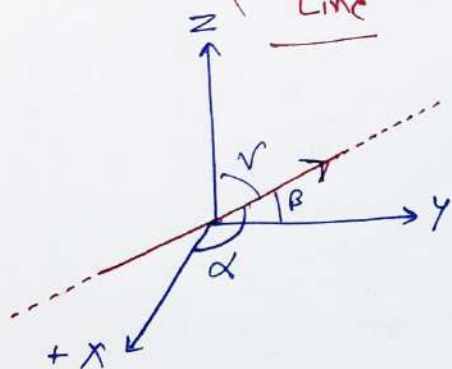
$$l^2 + m^2 + n^2 = 1$$

$$\lambda \in \mathbb{R} - \{0\}$$

Note: There are two possibilities of Direction Cosines (D.C.)

for the same line.

(Given Directed Line)



Direction Angles  $\rightarrow \alpha, \beta, \gamma$

D.C.  $\cos \alpha, \cos \beta, \cos \gamma$

(Given)

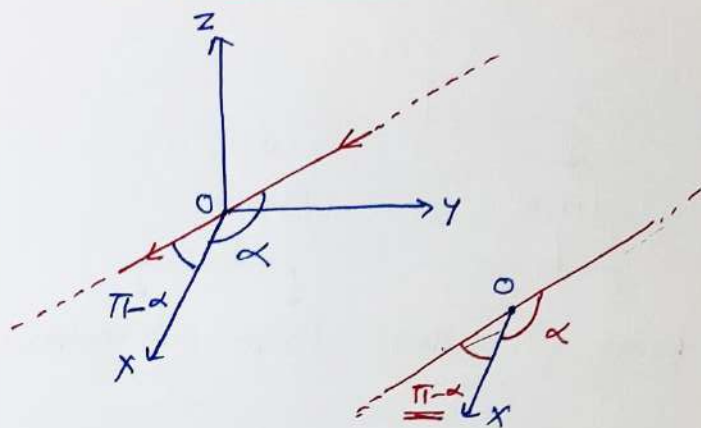
$\cos \alpha, \cos \beta, \cos \gamma$

Preferred

D.C.  $-\cos \alpha, -\cos \beta, -\cos \gamma$

possible but not preferred

(Reversed Line)



Direction Angles  $\rightarrow \pi - \alpha, \pi - \beta, \pi - \gamma$

D.C.  $\rightarrow \cos(\pi - \alpha), \cos(\pi - \beta), \cos(\pi - \gamma)$   
 $\rightarrow \underline{-\cos \alpha, -\cos \beta, -\cos \gamma}$



# Conversion

D.C.



D.R.

$$l, m, n \rightarrow \lambda l, \lambda m, \lambda n$$

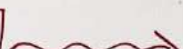
$$\cos \alpha, \cos \beta, \cos \gamma \rightarrow \lambda \cos \alpha, \lambda \cos \beta, \lambda \cos \gamma$$

$$\lambda \in \mathbb{R} - \{0\}$$

D.R.

$$\left\{ \begin{array}{l} \lambda=1, \quad \underline{l, m, n} \\ \lambda=2, \quad \underline{2l, 2m, 2n} \\ \lambda=3, \quad \underline{3l, 3m, 3n} \\ \vdots \quad \quad \quad \vdots \end{array} \right\}$$

D.R.



D.C.

$$a, b, c \rightarrow \frac{\pm a}{\sqrt{a^2+b^2+c^2}}, \frac{\pm b}{\sqrt{a^2+b^2+c^2}}, \frac{\pm c}{\sqrt{a^2+b^2+c^2}}$$

Note: A line Always has

unique D.C.

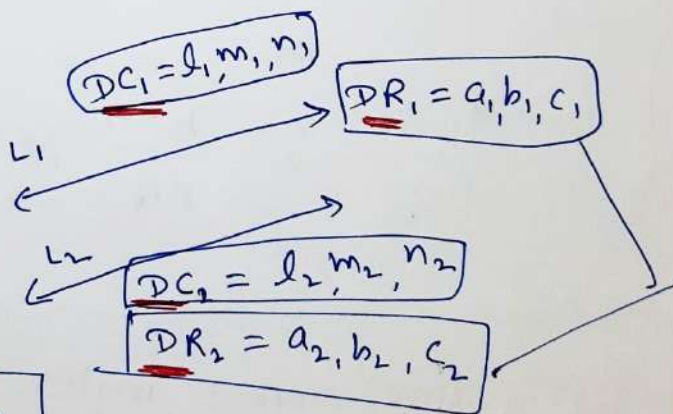
Infinite number of D.R.

Note: Parallel lines:

$$L_1 \parallel L_2$$

①  $DC_1 = DC_2$  (Equal)

$$\boxed{l_1 = l_2 \quad m_1 = m_2 \quad n_1 = n_2}$$



②  $DR_1 \propto DR_2$

(Proportion)

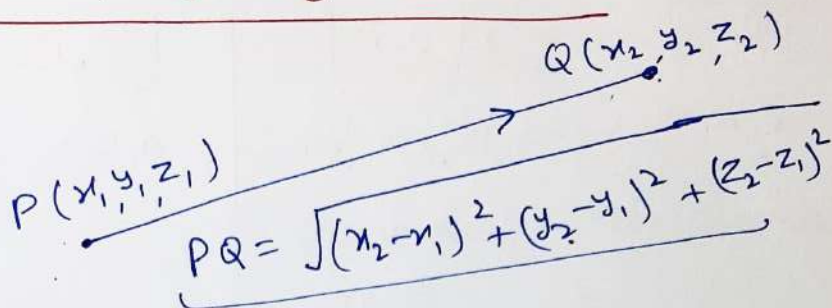
↓  
(Ratio → equal)

$$\boxed{\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}}$$

or

$$\boxed{a_1 : b_1 : c_1 = a_2 : b_2 : c_2}$$

D.R. & D.C. of a line joining two points.



$P(x_1, y_1, z_1)$   $Q(x_2, y_2, z_2)$

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

D.R.

$$\rightarrow \begin{cases} x_2 - x_1, y_2 - y_1, z_2 - z_1 \\ \text{or} \\ K(x_2 - x_1), K(y_2 - y_1), K(z_2 - z_1) \end{cases}$$

D.C.

$$\left\{ \begin{array}{l} \frac{(x_2 - x_1)}{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}}, \frac{(y_2 - y_1)}{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}}, \frac{(z_2 - z_1)}{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}} \\ \text{or} \\ \frac{x_2 - x_1}{PQ}, \frac{y_2 - y_1}{PQ}, \frac{z_2 - z_1}{PQ} \end{array} \right.$$



e.g. If a line makes  $\overbrace{90^\circ, 60^\circ, 30^\circ}$  with the positive direction of x, y & z-axis respectively, find its

Direction Cosines.

$$\begin{aligned} \rightarrow D.C. &= \cos \alpha, \cos \beta, \cos \gamma \\ &= \cos 90^\circ, \cos 60^\circ, \cos 30^\circ \\ &= 0, \frac{1}{2}, \frac{\sqrt{3}}{2} \end{aligned}$$

$$\begin{array}{ccc} \alpha, & \beta, & \gamma \\ \downarrow & \downarrow & \downarrow \\ 90^\circ & 60^\circ & 30^\circ \end{array}$$

e.g. If a line has direction ratios 2, -1, -2, determine its direction cosines.

$$\begin{aligned} \boxed{D.R. = 2, -1, -2} \\ \left[ \begin{aligned} D.C. &= \frac{2}{\sqrt{(2)^2 + (-1)^2 + (-2)^2}}, \frac{-1}{\sqrt{4+1+4}}, \frac{-2}{\sqrt{4+1+4}} \\ D.C. &= \frac{2}{3}, \frac{-1}{3}, \frac{-2}{3} \end{aligned} \right. \end{aligned}$$

e.g. Find the direction cosines of the line passing through the two points  $(-2, 4, -5)$  &  $(1, 2, 3)$

$$PQ = \sqrt{3^2 + (-2)^2 + 8^2}$$

$$= \sqrt{9 + 4 + 64}$$

$$= \sqrt{77}$$

$$D.R. = x_2 - x_1, y_2 - y_1, z_2 - z_1$$

$$= \underline{\underline{3, -2, 8}}$$

$$D.C. = \frac{3}{\sqrt{77}}, \frac{-2}{\sqrt{77}}, \frac{8}{\sqrt{77}}$$

e.g. Show that the points  $A(2, 3, -4)$ ,  $B(1, -2, 3)$  &  $C(3, 8, -11)$  are collinear.

in a line

$DC \& DP$



D.R.  $AB = ?$

D.R.  $BC = ?$

Proportion  
(Ratio equal)

$AB + BC = CA$

D.R.  $AB = 1-2, -2-3, 3-(-4) = -1, -5, 7$   
 $(2, 3, -4) \rightarrow (1, -2, 3)$

$-1 : -5 : 7$

$1 : 5 : -7$

D.R.  $BC = 3-1, 8-(-2), -11-3 = 2, 10, -14$   
 $(1, -2, 3) \rightarrow (3, 8, -11)$

$2 : 10 : -14$

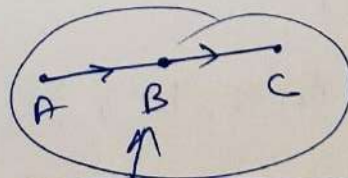
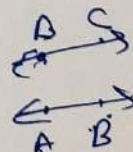
$1 : 5 : -7$

$\frac{-1}{2} = \frac{-5}{10} = \frac{7}{-14}$

$\frac{-1}{2} = \frac{-1}{2} = \frac{-1}{2}$

Collinear

$AB \parallel BC$



Common Point



## Exercise - 11.1

D.C. ★  
D.R.

Q.1 If a line makes angles  $90^\circ$ ,  $135^\circ$ ,  $45^\circ$  with the  $x$ ,  $y$  and  $z$ -axes respectively, find its direction cosines.

Direction Angles

 $\alpha, \beta, \gamma$   
 $\downarrow \quad \downarrow \quad \downarrow$   
 $90^\circ, 135^\circ, 45^\circ$ 

Direction Cosines

$$\begin{aligned} & \cos \alpha, \cos \beta, \cos \gamma \\ &= \cos(90^\circ), \cos(135^\circ), \cos(45^\circ) \\ &= 0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \end{aligned}$$

Q.2 Find the direction cosines of a line which makes equal angles with the coordinate axes.

$$\alpha = \beta = \gamma \quad \text{D.C.} \quad [\cos \alpha, \cos \beta, \cos \gamma] \rightsquigarrow \cos \alpha, \cos \alpha, \cos \alpha$$

Properties.

$$\Rightarrow \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\Rightarrow \cos^2 \alpha + \cos^2 \alpha + \cos^2 \alpha = 1$$

$$3 \cos^2 \alpha = 1 \Rightarrow \cos^2 \alpha = \frac{1}{3}$$

$$\Rightarrow \cos \alpha = \pm \frac{1}{\sqrt{3}}$$

$$\text{D.C.} \quad \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}$$

Q.3 If a line has the direction ratios  $-18, 12, -4$ , then what are its direction cosines?

$$\text{D.R.} \rightarrow -18, 12, -4$$

$$\begin{aligned} \text{D.C.} & \rightarrow \frac{-18}{\sqrt{(-18)^2 + (12)^2 + (-4)^2}}, \frac{12}{\sqrt{324 + 144 + 16}}, \frac{-4}{\sqrt{484}} \\ &= \frac{-18}{22}, \frac{12}{22}, \frac{-4}{22} = \left( \frac{-9}{11}, \frac{6}{11}, \frac{-2}{11} \right) \end{aligned}$$



Points

on a line

Q.4 Show that the  $(2,3,4)$ ,  $(-1,-2,1)$ ,  $(5,8,7)$  are collinear.



DR (Direction Ratios)

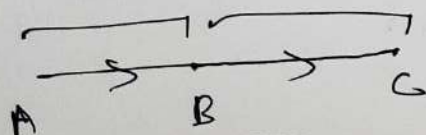
$$DR_{AB} = -1-2, -2-3, 1-4 \\ = -3, -5, -3$$

$$DR_{BC} = 5-(-1), 8-(-2), 7-1 \\ = 6, 10, 6$$

| $DR_{AB}$    | $DR_{BC}$  |
|--------------|------------|
| $-3, -5, -3$ | $6, 10, 6$ |
| $-3:-5:-3$   | $6:10:6$   |
| $3:5:3$      | $3:5:3$    |

Ratio  $\rightarrow$  equal.

$\widehat{DR} \rightarrow$  proportional  $\rightarrow$  parallel



Common

$A, B, C \rightarrow$  Collinear

Method-I

$$AB + BC = AC$$

Method-II

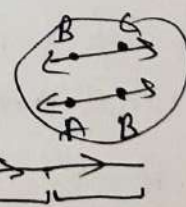
$$\text{ar}(\triangle ABC) = 0$$

Method-III

$DR, DC$

$$\frac{-3}{6} = \frac{-5}{10} = \frac{-3}{6} \\ \frac{-1}{2} = \frac{-1}{2} = \frac{-1}{2}$$

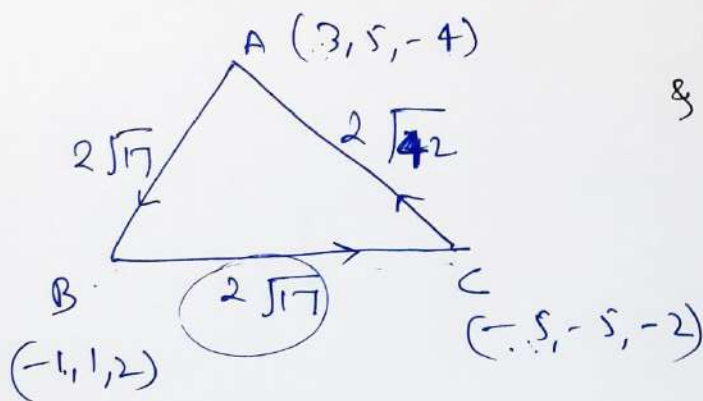
$AB \parallel BC$



**Q.5** Find the direction cosines of the sides of the triangle whose vertices are  $(3, 5, -4) \rightarrow A$

$$(-1, 1, 2) \rightarrow B$$

$$\& (-5, -5, -2) \rightarrow C$$



$$DC_{AB} = \frac{-1-3}{2\sqrt{17}}, \frac{1-5}{2\sqrt{17}}, \frac{2+4}{2\sqrt{17}}$$

$$= \frac{-4}{2\sqrt{17}}, \frac{-4}{2\sqrt{17}}, \frac{6}{2\sqrt{17}}$$

$$\rightarrow = -\frac{2}{\sqrt{17}}, -\frac{2}{\sqrt{17}}, \frac{3}{\sqrt{17}} \checkmark$$

$$DC_{BC} = \frac{-4}{2\sqrt{17}}, \frac{-6}{2\sqrt{17}}, \frac{-4}{2\sqrt{17}}$$

$$= -\frac{2}{\sqrt{17}}, -\frac{3}{\sqrt{17}}, -\frac{2}{\sqrt{17}}$$

$$DC_{CA} = \frac{8}{2\sqrt{42}}, \frac{10}{2\sqrt{42}}, \frac{-2}{2\sqrt{42}}$$

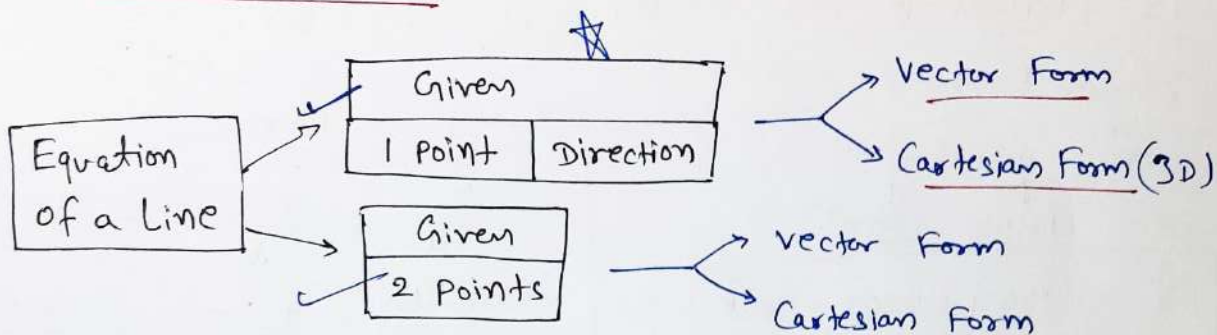
$$= \frac{4}{\sqrt{42}}, \frac{5}{\sqrt{42}}, -\frac{1}{\sqrt{42}}$$

$$\begin{aligned} AB &= \sqrt{4^2 + 4^2 + (-6)^2} \\ &= \sqrt{68} \\ &= 2\sqrt{17} \end{aligned}$$

$$\begin{aligned} BC &= \sqrt{(-4)^2 + (-6)^2 + (-4)^2} \\ &= \sqrt{68} \\ &= 2\sqrt{17} \end{aligned}$$

$$\begin{aligned} CA &= \sqrt{8^2 + 10^2 + (-2)^2} \\ &= \sqrt{64 + 100 + 4} \\ &= \sqrt{168} \\ &= 2\sqrt{42} \end{aligned}$$



Line (in 3D)Equation of a Line (when 1 point & Direction are given)

Let the line passes through the point A & is parallel to  $\vec{b}$

Coordinates of A ( $x_1, y_1, z_1$ )

Position Vector ( $\vec{a}$ )

of A =  $x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$  (Point)

$\vec{b} = a\hat{i} + b\hat{j} + c\hat{k}$  (Direction)

Equation of Line (L)  $\Rightarrow$

Vector Form

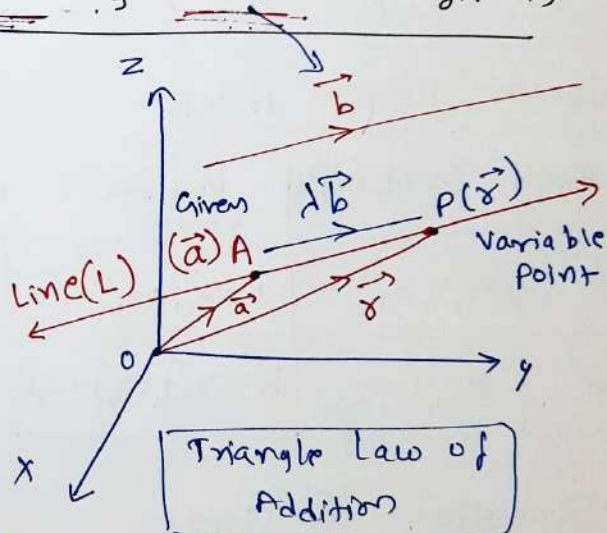
$\vec{r} = \vec{a} + \lambda \vec{b}$

Point Direction

Cartesian Form

$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$

Direction Ratios



$\vec{r} = \vec{a} + \lambda \vec{b}$

variable Point Direction

$(x\hat{i} + y\hat{j} + z\hat{k})$

$(x_1\hat{i} + y_1\hat{j} + z_1\hat{k})$

$a\hat{i} + b\hat{j} + c\hat{k}$

Comparison

Solve

e.g. Find the vector & Cartesian form of the line passing through the point  $(100, 200, 300)$  and which is parallel to the vector  $7\hat{i} - 17\hat{j} + 13\hat{k}$ .

Vector form,

$$\vec{r} = \vec{a} + \lambda \vec{b}$$

↑ Point
 ↑ Dire<sup>n</sup>.

$$\vec{r} = (100\hat{i} + 200\hat{j} + 300\hat{k}) + \lambda(7\hat{i} - 17\hat{j} + 13\hat{k})$$

Cartesian form,

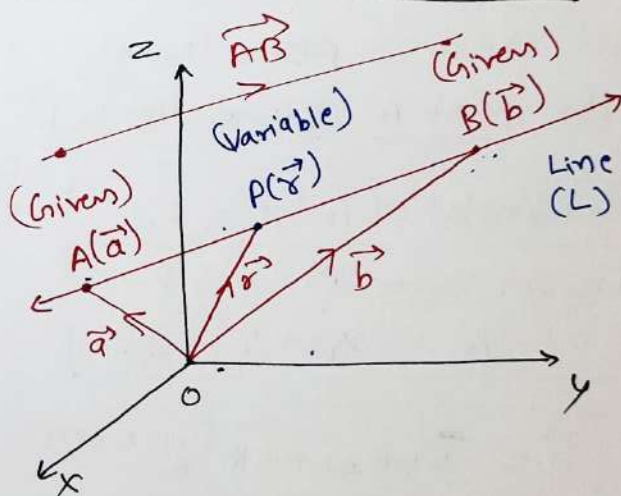
$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$$

$$\Rightarrow \frac{x-100}{7} = \frac{y-200}{-17} = \frac{z-300}{13}$$

Equation of a line (when two points on the line are given) →

Given Points A & B

| Point | Coordinates       | Position vector                                  |
|-------|-------------------|--------------------------------------------------|
| A     | $(x_1, y_1, z_1)$ | $\vec{a} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$ |
| B     | $(x_2, y_2, z_2)$ | $\vec{b} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$ |



Direction of line.

$$\vec{AB} = \vec{b} - \vec{a} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$$

Equation of Line(L)

Vector Form

$$\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$$

↑ Point
 ↑ Direction

Cartesian Form

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

↑ Direction.



e.g. Find the vector equation for the line passing through the points  $(7, -17, 27)$  &  $(-1, 2, -3)$ .

Vector form.

$$\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$$

Point                      Direction

$$\vec{r} = 7\hat{i} - 17\hat{j} + 27\hat{k} + \lambda(-8\hat{i} + 19\hat{j} - 30\hat{k})$$

Cartesian form.

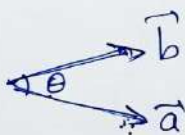
$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

$$\Rightarrow \frac{x-7}{-8} = \frac{y+17}{19} = \frac{z-27}{-30}$$

Angle between two lines :

Dot Product

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$



understand in Cartesian Form

$$\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1} \rightarrow L_1$$

$$\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2} \rightarrow L_2$$

$\theta$

Understand in Vector Form

$$\vec{r} = \vec{a}_1 + \lambda \vec{b}_1 \quad (L_1)$$

$$\vec{r} = \vec{a}_2 + \mu \vec{b}_2 \quad (L_2)$$

$\theta$

Directly by Dot Product.

$$\cos \theta = \frac{\vec{b}_1 \cdot \vec{b}_2}{|\vec{b}_1| |\vec{b}_2|}$$

Conditions

Case-I perpendicular lines

$$\theta = 90^\circ$$

$$a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$$

Case-II Parallel lines

$$\theta = 0^\circ$$

$$\vec{DR}_1 \parallel \vec{DR}_2$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\vec{DR}_1 \propto \vec{DR}_2$$

$$\cos 90^\circ = 0$$

e.g. Find the angle between the pair of lines given by

$$\vec{r} = \underline{3\hat{i} + 2\hat{j} - 4\hat{k}} + \lambda(\underline{\hat{i} + 2\hat{j} + 2\hat{k}}) \quad \vec{b}_1 \quad \text{Directions}$$

$$\vec{r} = \underline{5\hat{i} - 2\hat{j}} + \mu(\underline{3\hat{i} + 2\hat{j} + 6\hat{k}}) \quad \vec{b}_2$$

$$\cos\theta = \frac{\vec{b}_1 \cdot \vec{b}_2}{|\vec{b}_1| |\vec{b}_2|}$$

$$\cos\theta = \frac{(\hat{i} + 2\hat{j} + 2\hat{k}) \cdot (3\hat{i} + 2\hat{j} + 6\hat{k})}{\sqrt{1^2 + 2^2 + 2^2} \sqrt{3^2 + 2^2 + 6^2}} = \frac{19}{21}$$

$$\theta = \cos^{-1}\left(\frac{19}{21}\right)$$

e.g. Find the angle between the pair of lines

$$\frac{x+3}{\underline{3}} = \frac{y-1}{\underline{5}} = \frac{z+3}{\underline{4}} \quad \& \quad \frac{x+1}{\underline{1}} = \frac{y-4}{\underline{1}} = \frac{z-5}{\underline{2}}$$

Directions

$$\cos\theta = \frac{(3 \times 1) + (5 \times 1) + (4 \times 2)}{\sqrt{3^2 + 5^2 + 4^2} \cdot \sqrt{1^2 + 1^2 + 2^2}} = \frac{3 + 5 + 8}{\sqrt{50} \cdot \sqrt{6}}$$

$$\cos\theta = \frac{16}{\sqrt{300}} = \frac{16}{\sqrt{100 \times 3}} = \frac{16}{10\sqrt{3}} = \frac{8}{5\sqrt{3}}$$

~~150 = 25 \times 6~~

$$\theta = \cos^{-1}\left(\frac{8}{5\sqrt{3}}\right)$$



## Distance between Two Skew Lines

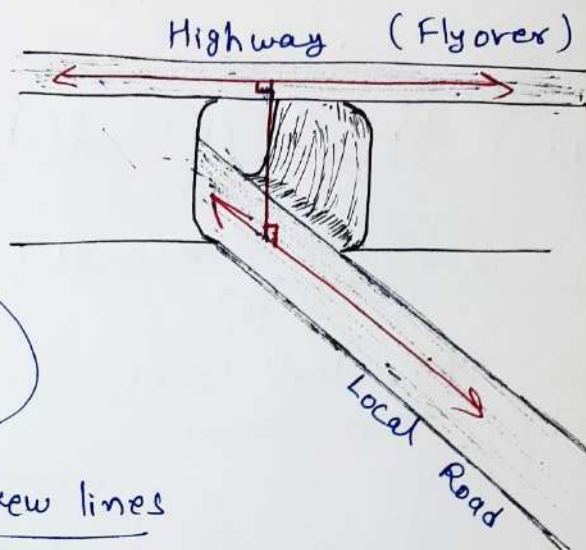
Non Coplanar  
(एक plane में हो ही नहीं सकती)

Not Parallel  
Never meet

### Skew Lines

$$\vec{r} = \vec{a}_1 + \lambda \vec{b}_1 \quad (L_1)$$

$$\vec{r} = \vec{a}_2 + \mu \vec{b}_2 \quad (L_2)$$



### ★ Shortest Distance b/w Skew lines

$$d = \left| \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|} \right|$$

↑  
modulus  
magnitude.

Remember only  
one formula  
(vector form only)

### Distance b/w Parallel lines

$$L_1 \quad \vec{r} = \vec{a}_1 + \lambda \vec{b}$$

$$L_2 \quad \vec{r} = \vec{a}_2 + \mu \vec{b}$$

$$L_1 \parallel L_2$$

$$d = \left| \frac{(\vec{a}_2 - \vec{a}_1) \times \vec{b}}{|\vec{b}|} \right|$$

e.g. Find the shortest distance between the lines  $l_1$  &  $l_2$

given by  $\vec{r} = \hat{i} + 2\hat{j} - 4\hat{k} + \lambda(\hat{i} - \hat{j} + 2\hat{k}) \leftarrow l_1$

&  $\vec{r} = \hat{i} + 2\hat{j} + \mu(-\hat{i} + \hat{j} - 2\hat{k}) \leftarrow l_2$

$l_1: \vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(\hat{i} - \hat{j} + 2\hat{k})$

$\vec{a}_1$   $\vec{b}$

$l_2: \vec{r} = (\hat{i} + 2\hat{j}) + \mu(-\hat{i} + \hat{j} - 2\hat{k})$

$\vec{a}_2$

Parallel lines.

~~$\mu$~~   $-\mu$   $\cdot \lambda_2$

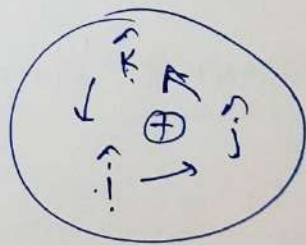
$\vec{r} = \vec{a}_1 + \lambda \vec{b}$

$\vec{r} = \vec{a}_2 + \mu \vec{b}$

$d$

$d = \left| \frac{(\vec{a}_2 - \vec{a}_1) \times \vec{b}}{|\vec{b}|} \right| = \left| \frac{(0\hat{i} + 0\hat{j} + 4\hat{k}) \times (\hat{i} - \hat{j} + 2\hat{k})}{\sqrt{1^2 + (-1)^2 + (2)^2}} \right|$

$d = \left| \frac{4(\hat{k} \times \hat{i}) - 4(\hat{k} \times \hat{j}) + 8(0)}{\sqrt{1+1+4}} \right|$



$d = \left| \frac{4\hat{j} + 4\hat{i}}{\sqrt{6}} \right| = \frac{\sqrt{16+16}}{\sqrt{6}} = \frac{\sqrt{2+16}}{\sqrt{2 \times 3}}$

$d = \frac{4}{\sqrt{3}}$



e.g. Find the shortest distance between the lines

$l_1$  &  $l_2$  whose vector equations are

$$\vec{r} = (\hat{i} + \hat{j}) + \lambda(2\hat{i} - \hat{j} + \hat{k}) \quad \& \quad \vec{b}_1$$

$$\vec{r} = (2\hat{i} + \hat{j} - \hat{k}) + \mu(3\hat{i} - 5\hat{j} + 2\hat{k})$$

Skew lines

$$\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$$

$$\vec{r} = \vec{a}_2 + \mu \vec{b}_2$$

$$d = \left| \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|} \right|$$

$$\vec{b}_1 \times \vec{b}_2 = (2\hat{i} - \hat{j} + \hat{k}) \times (3\hat{i} - 5\hat{j} + 2\hat{k})$$

Cross Product

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 1 \\ 3 & -5 & 2 \end{vmatrix} = \hat{i}(-2+5) - \hat{j}(4-3) + \hat{k}(-10+3)$$

$$\vec{b}_1 \times \vec{b}_2 = 3\hat{i} - \hat{j} - 7\hat{k}$$

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{9+1+49} = \sqrt{59}$$

$$d = \left| \frac{(\hat{i} - \hat{k}) \cdot (3\hat{i} - \hat{j} - 7\hat{k})}{\sqrt{59}} \right| = \left| \frac{3+7}{\sqrt{59}} \right|$$

$$d = \frac{10}{\sqrt{59}}$$

## Exercise 11.2

Line (3D)



$$\vec{r} = \vec{a} + \lambda \vec{b}$$

Point

Direction.

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$$

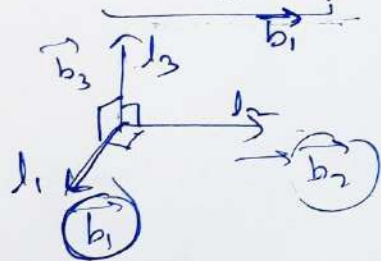
Q.1 Show that the three lines with direction cosines

$$\frac{12}{13}, \frac{-3}{13}, \frac{-4}{13};$$

$$\frac{4}{13}, \frac{12}{13}, \frac{3}{13};$$

$$\frac{3}{13}, \frac{-4}{13}, \frac{12}{13}$$

are mutually perpendicular.



$$\vec{b}_1 = \frac{12}{13} \hat{i} - \frac{3}{13} \hat{j} - \frac{4}{13} \hat{k}$$

$$\vec{b}_2 = \frac{4}{13} \hat{i} + \frac{12}{13} \hat{j} + \frac{3}{13} \hat{k}$$

$$\vec{b}_3 = \frac{3}{13} \hat{i} - \frac{4}{13} \hat{j} + \frac{12}{13} \hat{k}$$

$$\vec{b}_1 \cdot \vec{b}_2 = \left( \frac{12}{13} \hat{i} - \frac{3}{13} \hat{j} - \frac{4}{13} \hat{k} \right) \cdot \left( \frac{4}{13} \hat{i} + \frac{12}{13} \hat{j} + \frac{3}{13} \hat{k} \right)$$

$$= \frac{12 \times 4}{13 \times 13} - \frac{3 \times 12}{13 \times 13} - \frac{4 \times 3}{13 \times 13} = \frac{48 - 36 - 12}{13 \times 13}$$

$$= 0$$

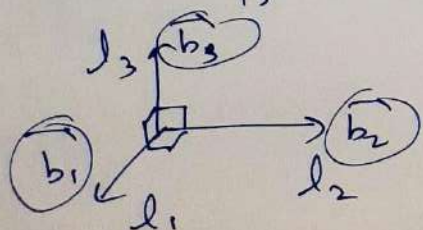
$$\vec{b}_1 \cdot \vec{b}_2 = 0$$

$$\vec{b}_1 \perp \vec{b}_2$$

$$l_1 \perp l_2$$

$$\vec{b}_2 \cdot \vec{b}_3 = \frac{4 \times 3}{13^2} - \frac{12 \times 4}{13^2} + \frac{3 \times 12}{13^2} = \frac{12 - 48 + 36}{13^2} = 0$$

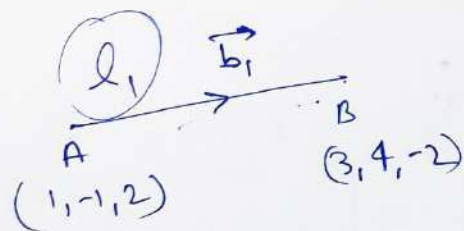
$$\vec{b}_1 \cdot \vec{b}_3 = \frac{12 \times 3}{13^2} + \frac{3 \times 4}{13^2} - \frac{4 \times 12}{13^2} = \frac{36 + 12 - 48}{13^2} = 0$$



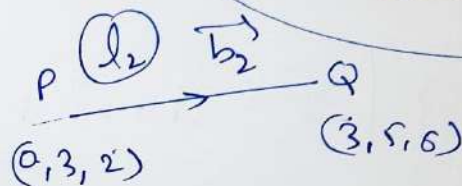
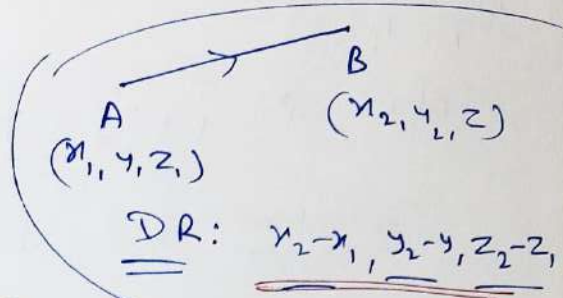
$$l_1 \perp l_2, l_2 \perp l_3, l_3 \perp l_1$$



**Q.2** Show that the line through the points  $(1, -1, 2)$ ,  $(3, 4, -2)$  is perpendicular to the line through the point  $(0, 3, 2)$  and  $(3, 5, 6)$ .

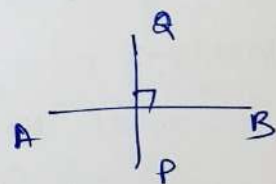
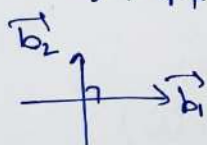


$$\begin{aligned} DR_1 &= 3-1, 4-(-1), -2-2 \\ &= 2, 5, -4 \\ \vec{b}_1 &= 2\hat{i} + 5\hat{j} - 4\hat{k} \end{aligned}$$

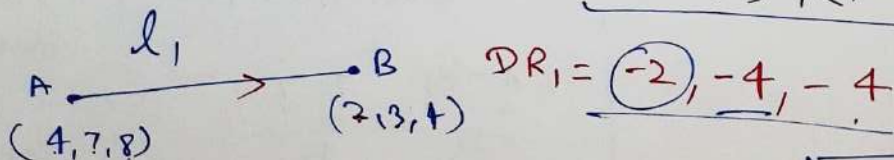


$$\begin{aligned} DR_2 &= 3, 2, 4 \\ \vec{b}_2 &= 3\hat{i} + 2\hat{j} + 4\hat{k} \end{aligned}$$

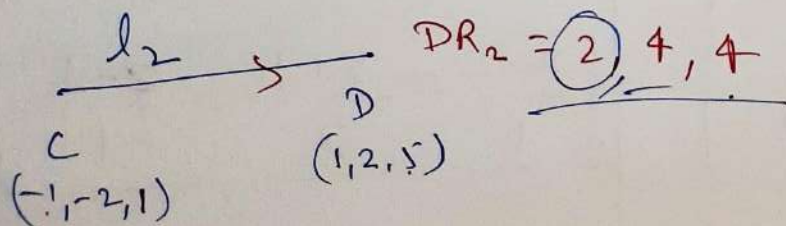
$$\begin{aligned} \vec{b}_1 \cdot \vec{b}_2 &= (2\hat{i} + 5\hat{j} - 4\hat{k}) \cdot (3\hat{i} + 2\hat{j} + 4\hat{k}) \\ &= 6 + 10 - 16 \\ &= 16 - 16 = 0 \end{aligned}$$



**Q.3** Show that the line through the points  $(4, 7, 8)$ ,  $(2, 3, 4)$  is parallel to the line through the points  $(-1, -2, 1)$ ,  $(1, 2, 5)$ .



$$DR_1 = -2, -4, -4$$



$$DR_2 = 2, 4, 4$$

$$\frac{-2}{2} = \frac{-4}{4} = \frac{-4}{4}$$

$$-1 = -1 = -1$$

DRs are proportional

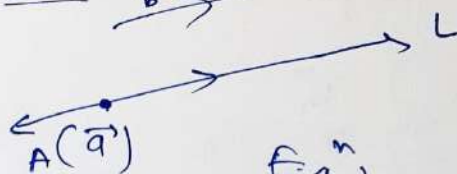
$\therefore$  lines are  $\parallel$ .

[Q.4] Find the Equation of the line which passes through the point  $(1, 2, 3)$  and is parallel to the vector  $(3\hat{i} + 2\hat{j} - 2\hat{k}) = \vec{b}$

Point  
A(1, 2, 3)

$$A(\vec{a}) \quad \vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$$

Direction,  $\vec{b} = 3\hat{i} + 2\hat{j} - 2\hat{k}$



$$\vec{r} = \vec{a} + \lambda \vec{b}$$

Varying Point Direction

$$\Rightarrow \vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(3\hat{i} + 2\hat{j} - 2\hat{k})$$

[Q.5] Find the Equation of the line in vector and in Cartesian form that passes through the point with position vector  $2\hat{i} - \hat{j} + 4\hat{k}$  and is in the direction of  $\hat{i} + 2\hat{j} - \hat{k}$ .

Point (2, -1, 4)

Direction ( $\vec{b}$ )

D.R. = 1, 2, -1

Eq<sup>n</sup>. of line

Vector form,

$$\vec{r} = \vec{a} + \lambda \vec{b}$$

Point      Dir.

$$\vec{r} = (2\hat{i} - \hat{j} + 4\hat{k}) + \lambda(\hat{i} + 2\hat{j} - \hat{k})$$

Cartesian form

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

Point

Dir.

$$\frac{x - 2}{1} = \frac{y + 1}{2} = \frac{z - 4}{-1}$$



Q.6 Find the Cartesian eq<sup>n</sup>. of the line which passes through the point  $(-2, 4, -5)$  and parallel to the line given by

$$\frac{x+3}{3} = \frac{y-4}{5} = \frac{z+8}{6}$$

Dir.  $3, 5, 6$

Dir.

?  $(-2, 4, -5)$

$$\frac{x - (-2)}{a} = \frac{y - 4}{b} = \frac{z - (-5)}{c}$$

3, 5, 6

$$\frac{x+2}{3} = \frac{y-4}{5} = \frac{z+5}{6}$$

Q.7 The Cartesian equation of a line is  $\frac{x-5}{3} = \frac{y+4}{7} = \frac{z-6}{2}$ . Write its Vector form

$$\vec{r} = \vec{a} + \lambda \vec{b}$$

Point Direction

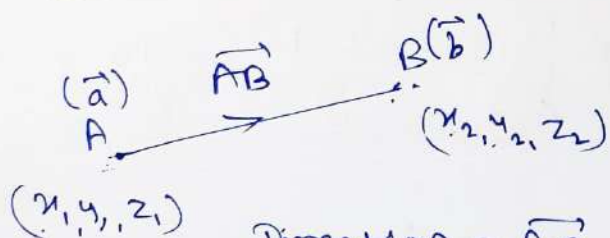
$(5, -4, 6)$  Point

$$\frac{x-5}{3} = \frac{y+4}{7} = \frac{z-6}{2}$$

Direction  $(3, 7, 2)$

$$\vec{r} = (5\hat{i} - 4\hat{j} + 6\hat{k}) + \lambda (3\hat{i} + 7\hat{j} + 2\hat{k})$$

Q.8 Find the Vector and the Cartesian equations of the line that passes through the origin  $(0,0,0)$  and  $(5,-2,3)$ .



$$\text{Direction} = \vec{AB} = \vec{b} - \vec{a}$$

1 Point, 1 Dir.

$$\vec{r} = \vec{a} + \lambda \vec{b}$$

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$$

Eqn. of line,

Vector form,

$$\vec{r} = \vec{a} + \lambda (\vec{b} - \vec{a})$$

Cartesian form,

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

ATQ A ( $\vec{a}$ )  $(0,0,0)$   
B ( $\vec{b}$ )  $(5,-2,3)$

Vector Form

$$\vec{r} = (0\hat{i} + 0\hat{j} + 0\hat{k}) + \lambda (5\hat{i} - 2\hat{j} + 3\hat{k})$$

$$\vec{r} = \lambda (5\hat{i} - 2\hat{j} + 3\hat{k})$$

Cartesian Form

$$\frac{x-0}{5-0} = \frac{y-0}{-2-0} = \frac{z-0}{3-0}$$

$$\Rightarrow \frac{x}{5} = \frac{y}{-2} = \frac{z}{3}$$

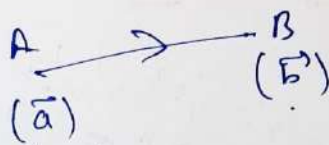


**Q.9** Find the vector and the Cartesian Equations of the line that passes through the points

$$(3, -2, -5), (3, -2, 6).$$

A

B



$$(x_1, y_1, z_1) \quad (x_2, y_2, z_2)$$

Cartesian form,

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

$$\Rightarrow \frac{x-3}{3-3} = \frac{y-(-2)}{-2+2} = \frac{z-(-5)}{6+5}$$

$$\Rightarrow \boxed{\frac{x-3}{0} = \frac{y+2}{0} = \frac{z+5}{11}}$$

Vector form

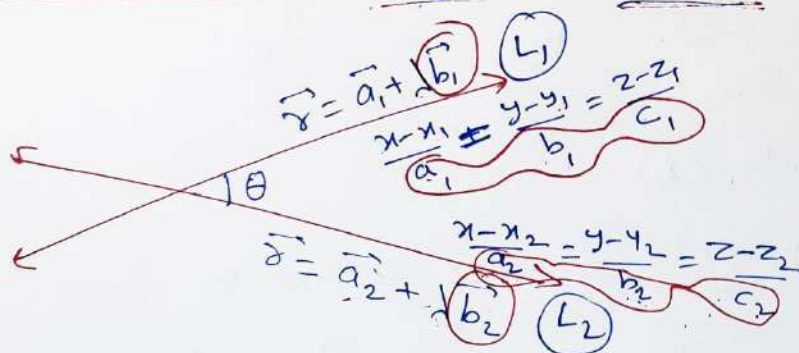
$$\vec{r} = \underbrace{(\vec{a})}_{\text{Point A}} + \lambda(\vec{b}-\vec{a})$$

$$\Rightarrow \vec{r} = 3\hat{i} - 2\hat{j} - 5\hat{k} + \lambda((3-3)\hat{i} + (-2+2)\hat{j} + (6+5)\hat{k})$$

$$\rightarrow \vec{r} = (3\hat{i} - 2\hat{j} - 5\hat{k}) + \lambda(0\hat{i} + 0\hat{j} + 11\hat{k})$$

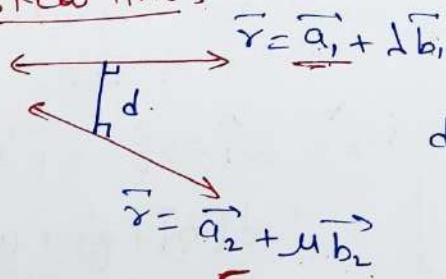
$$\boxed{\vec{r} = (3\hat{i} - 2\hat{j} - 5\hat{k}) + \lambda(11\hat{k})}$$

## Exercise 11.2

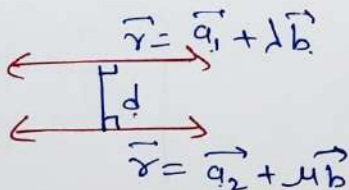
(Angle & Distance b/w Two lines)

$$\cos \theta = \frac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1| |\vec{b}_2|}$$

$$\cos \theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Skew lines

$$d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$$

Parallel lines

$$d = \frac{|(\vec{a}_2 - \vec{a}_1) \times \vec{b}|}{|\vec{b}|}$$

Q.10 Find the angle b/w the following pairs of lines:

$$(i) \quad \vec{r} = 2\hat{i} - 5\hat{j} + \hat{k} + \lambda(3\hat{i} + 2\hat{j} + 6\hat{k}) \quad \vec{b}_1 = 3\hat{i} + 2\hat{j} + 6\hat{k}$$

$$\vec{r} = 7\hat{i} - 6\hat{k} + \mu(\hat{i} + 2\hat{j} + 2\hat{k}) \quad \vec{b}_2 = \hat{i} + 2\hat{j} + 2\hat{k}$$

$$\cos \theta = \frac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1| |\vec{b}_2|} = \frac{(3\hat{i} + 2\hat{j} + 6\hat{k}) \cdot (\hat{i} + 2\hat{j} + 2\hat{k})}{\sqrt{9+4+36} \cdot \sqrt{1^2+4+4}}$$

$$\cos \theta = \frac{3+4+12}{7 \times 3} = \left(\frac{19}{21}\right) \Rightarrow \theta = \cos^{-1}\left(\frac{19}{21}\right)$$



$$(ii) \vec{r} = (3\hat{i} + \hat{j} - 2\hat{k}) + \lambda(\hat{i} - \hat{j} - 2\hat{k}) \quad | \quad \vec{r} = (2\hat{i} - \hat{j} - 5\hat{k}) + \mu(3\hat{i} - 5\hat{j} - 4\hat{k})$$

$\vec{b}_1$   $\vec{b}_2$

$$\cos \theta = \frac{\vec{b}_1 \cdot \vec{b}_2}{|\vec{b}_1| |\vec{b}_2|} = \frac{3+5+8}{\sqrt{1+1+4} \sqrt{9+25+16}}$$

$$\cos \theta = \frac{16}{\sqrt{6} \sqrt{50}} = \frac{16}{\sqrt{300}} = \frac{16}{\sqrt{100 \times 3}} = \frac{16}{10\sqrt{3}} = \frac{8}{5\sqrt{3}}$$

$$\cos \theta = \frac{8}{5\sqrt{3}} \Rightarrow \theta = \cos^{-1}\left(\frac{8}{5\sqrt{3}}\right)$$

[Q.1] Find the angle between the following pair of lines:

Cartesian form

$$(i) \frac{x-2}{2} = \frac{y-1}{5} = \frac{z+3}{-3} ; \frac{x+2}{-1} = \frac{y-4}{8} = \frac{z-5}{4}$$

$(a_1) \quad (b_1) \quad (c_1) \quad (a_2) \quad (b_2) \quad (c_2)$

$$\cos \theta = \left| \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}} \right| = \left| \frac{-2 + 40 - 12}{\sqrt{4+25+9} \sqrt{1+64+16}} \right|$$

$$\Rightarrow \cos \theta = \frac{26}{\sqrt{38} \sqrt{81}} = \frac{26}{9\sqrt{38}}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{26}{9\sqrt{38}}\right)$$

$$(ii) \frac{x}{2} = \frac{y}{2} = \frac{z}{1} \& \frac{x-5}{4} = \frac{y-2}{1} = \frac{z-3}{8}$$

$(a_1) \quad (b_1) \quad (c_1) \quad (a_2) \quad (b_2) \quad (c_2)$

$$\cos \theta = \left| \frac{8+2+8}{\sqrt{4+4+1} \sqrt{16+1+64}} \right| = \frac{18^2}{3 \times 81} = \frac{2}{3}$$

$$\theta = \cos^{-1}\left(\frac{2}{3}\right)$$

Q.12 Find the value of  $(P)$  so that the lines

$$\frac{1-x}{3} = \frac{7y-14}{2P} = \frac{z-3}{2} \quad \& \quad \frac{7-7x}{3P} = \frac{y-5}{1} = \frac{6-z}{5} \text{ are at Right angle.}$$

C.F.

Standard forms

$$\theta = 90^\circ$$

$$\cos 90^\circ = 0$$

$$\Rightarrow \frac{-(x-1)}{3} = \frac{7(y-2)}{2P} = \frac{z-3}{2}$$

$$\Rightarrow \frac{-7(x-1)}{3P} = \frac{y-5}{1} = \frac{-(z-6)}{5}$$

$$\Rightarrow \left[ \frac{x-1}{\textcircled{-3}} = \frac{y-2}{\textcircled{\frac{2P}{7}}} = \frac{z-3}{\textcircled{2}} \right]$$

$a_1 \quad b_1 \quad c_1$

$$\Rightarrow \left[ \frac{x-1}{\textcircled{\frac{3P}{-7}}} = \frac{y-5}{\textcircled{1}} = \frac{z-6}{\textcircled{-5}} \right]$$

$a_2 \quad b_2 \quad c_2$

$$\cos \theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

$$\theta = 90^\circ$$

$$\theta = 90^\circ, \cos \theta = \cos 90^\circ = \underline{0}$$

$$\Rightarrow 0 = \frac{(-3) \cdot \left(\frac{3P}{-7}\right) + \frac{2P}{7} + 2(-5)}{\sqrt{\quad} \sqrt{\quad}}$$

$$\Rightarrow 0 = \frac{9P}{7} + \frac{2P}{7} - 10$$

$$\Rightarrow 10 = \frac{11P}{7}$$

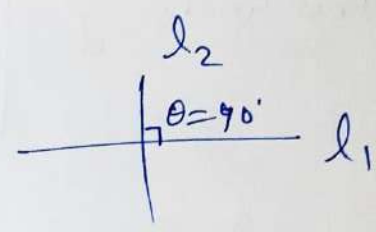
$$\Rightarrow \frac{70}{11} = P$$



**Q.13** Show that the lines  $\frac{x-5}{1} = \frac{y+2}{-5} = \frac{z}{1}$  &  $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$  are perpendicular to each other.

$$\cos \theta = \frac{7 + (-10) + (3)}{\sqrt{49+25+1} \cdot \sqrt{1+4+9}} = 0$$

$$\cos \theta = 0 \Rightarrow \theta = 90^\circ$$



**Q.14** Find the Shortest distance b/w the lines  
 $\vec{r} = (\hat{i} + 2\hat{j} + \hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$  &  $\vec{r} = (2\hat{i} - \hat{j} - \hat{k}) + \mu(2\hat{i} + \hat{j} + 2\hat{k})$   
 Point  $(\vec{a}_1)$  Direction  $(\vec{b}_1)$  Point  $(\vec{a}_2)$  Direction  $(\vec{b}_2)$

Skew lines

$$d = \left| \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|} \right|$$

$$d = \left| \frac{(\hat{i} - 3\hat{j} - 2\hat{k}) \cdot (-3\hat{i} + 3\hat{k})}{3\sqrt{2}} \right|$$

$$\Rightarrow d = \left| \frac{-3 - 6}{3\sqrt{2}} \right| = \frac{3}{\sqrt{2}}$$

$$\Rightarrow d = \frac{3}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$\Rightarrow d = \frac{3\sqrt{2}}{2}$$

$$\vec{b}_1 \times \vec{b}_2 = (\hat{i} - \hat{j} + \hat{k}) \times (2\hat{i} + \hat{j} + 2\hat{k})$$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 2 & 1 & 2 \end{vmatrix} \rightarrow \text{Expand}$$

$$= \hat{i}(-3) - \hat{j}(0) + \hat{k}(3) = -3\hat{i} + 3\hat{k}$$

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{9+9} = \sqrt{9 \times 2} = 3\sqrt{2}$$

Q.15 Find the Shortest distance b/w the lines

$$\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1}$$

Point  $\vec{a}_1 = -\hat{i} - \hat{j} - \hat{k}$  ✓

Direction  $\vec{b}_1 = 7\hat{i} - 6\hat{j} + \hat{k}$  ✓

$$\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$$

Point  $\vec{a}_2 = 3\hat{i} + 5\hat{j} + 7\hat{k}$  ✓

Direction  $\vec{b}_2 = \hat{i} - 2\hat{j} + \hat{k}$  ✓

Skew lines.

$$d = \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|}$$

$$d = \frac{(4\hat{i} + 6\hat{j} + 8\hat{k}) \cdot (-4\hat{i} - 6\hat{j} - 8\hat{k})}{2\sqrt{29}}$$

$$d = \frac{-16 - 36 - 64}{2\sqrt{29}}$$

$$d = \frac{+116}{2\sqrt{29}} = \frac{2 \times 29 \sqrt{29}}{2\sqrt{29}}$$

$$d = 2\sqrt{29}$$

$$\vec{b}_1 \times \vec{b}_2 = (7\hat{i} - 6\hat{j} + \hat{k}) \times (\hat{i} - 2\hat{j} + \hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & -6 & 1 \\ 1 & -2 & 1 \end{vmatrix}$$

$$= \hat{i}(-4) - \hat{j}(6) + \hat{k}(-8)$$

$$\vec{b}_1 \times \vec{b}_2 = -4\hat{i} - 6\hat{j} - 8\hat{k}$$

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{16 + 36 + 64}$$

$$= \sqrt{116}$$

$$= \sqrt{4 \times 29}$$

$$= 2\sqrt{29}$$



**Q.16** Find the shortest distance between the lines

whose vector equations are

$$\vec{r} = \underbrace{(\hat{i} + 2\hat{j} + 3\hat{k})}_{\text{Point } (\vec{a}_1)} + \lambda \underbrace{(\hat{i} - 3\hat{j} + 2\hat{k})}_{\text{Direction } (\vec{b}_1)}, \quad \vec{r} = \underbrace{(4\hat{i} + 5\hat{j} + 6\hat{k})}_{\text{Point } (\vec{a}_2)} + \mu \underbrace{(2\hat{i} + 3\hat{j} + \hat{k})}_{\text{Direction } (\vec{b}_2)}$$

Skew lines

$$d = \left| \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|} \right|$$

$$d = \left| \frac{(3\hat{i} + 3\hat{j} + 3\hat{k}) \cdot (-9\hat{i} + 3\hat{j} + 9\hat{k})}{3\sqrt{19}} \right|$$

$$d = \left| \frac{-27 + 9 + 27}{3\sqrt{19}} \right|$$

$$d = \frac{3}{\sqrt{19}}$$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & 2 \\ 2 & 3 & 1 \end{vmatrix} \rightarrow$$

$$= \hat{i}(-9) - \hat{j}(-3) + \hat{k}(9) \\ = -9\hat{i} + 3\hat{j} + 9\hat{k}$$

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{81 + 9 + 81}$$

$$= \sqrt{171}$$

$$= \sqrt{9 \times 19}$$

$$= 3\sqrt{19}$$

Q.17) Find the shortest distance b/w the lines whose vector equations are  $\vec{r} = (1-t)\hat{i} + (t-2)\hat{j} + (3-2t)\hat{k}$

$$\& \vec{r} = (s+1)\hat{i} + (2s-1)\hat{j} - (2s+1)\hat{k}.$$

⊥

$$L_1: \vec{r} = (1-t)\hat{i} + (t-2)\hat{j} + (3-2t)\hat{k}$$

$$L_1: \vec{r} = \underbrace{(\hat{i} - 2\hat{j} + 3\hat{k})}_{\text{Point } (\vec{a}_1)} + t \underbrace{(-\hat{i} + \hat{j} - 2\hat{k})}_{\text{Direction } (\vec{b}_1)}$$

Similarly

$$L_2: \vec{r} = \underbrace{(\hat{i} - \hat{j} - \hat{k})}_{\text{Point } (\vec{a}_2)} + s \underbrace{(\hat{i} + 2\hat{j} - 2\hat{k})}_{\text{Direction } (\vec{b}_2)}$$

Skew lines,

$$d = \left| \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|} \right|$$

$$d = \left| \frac{(\hat{j} - 4\hat{k}) \cdot (2\hat{i} - 4\hat{j} - 3\hat{k})}{\sqrt{29}} \right|$$

$$d = \left| \frac{-4 + 12}{\sqrt{29}} \right|$$

$$\boxed{d = \frac{8}{\sqrt{29}}}$$

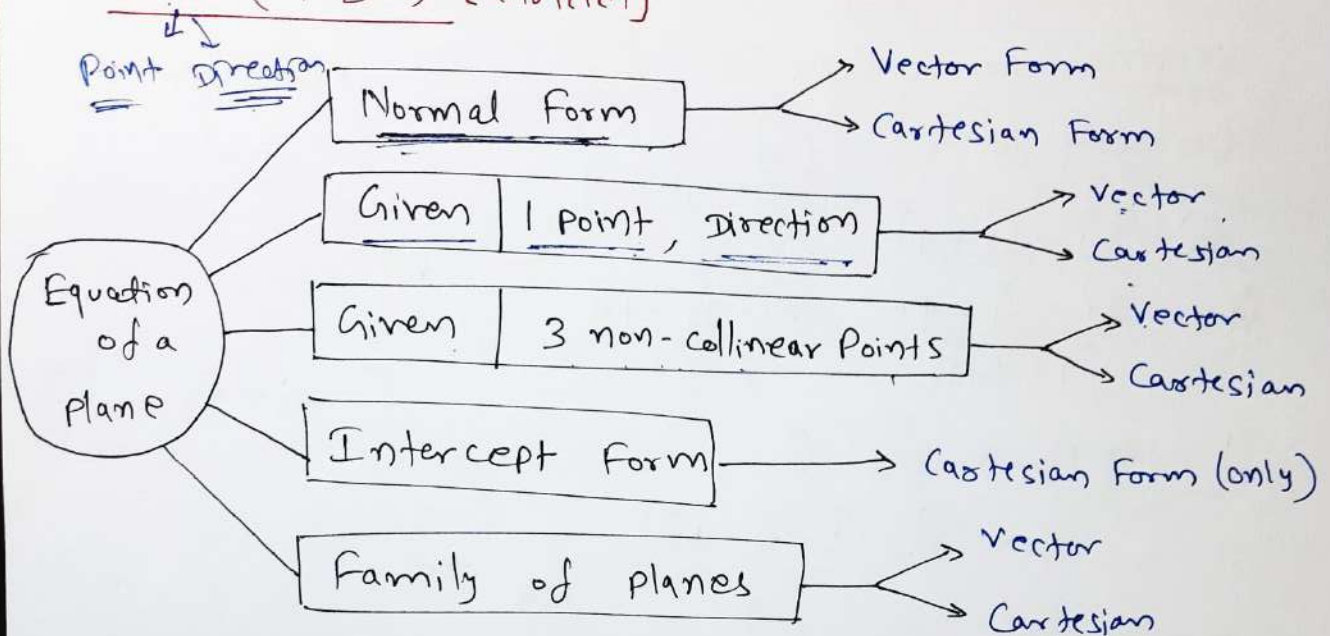
$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & -2 \\ 1 & 2 & -2 \end{vmatrix}$$

$$= \hat{i}(2) - \hat{j}(4) + \hat{k}(-3)$$

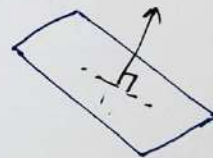
$$= 2\hat{i} - 4\hat{j} - 3\hat{k}$$

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{4 + 16 + 9} = \sqrt{29}$$

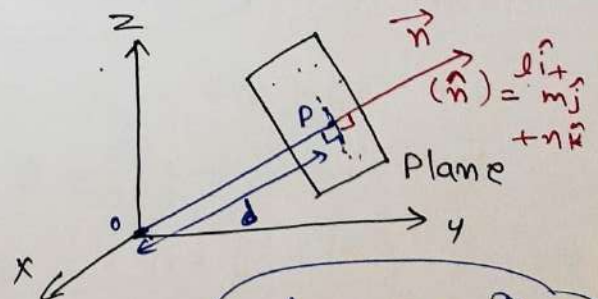


Plane (in 3D) [समतल]Note: Line में Direction का मतलब

(Line के along या parallel vector की Direction)

Plane में Direction का मतलब

(Plane के perpendicular vector की Direction)

Normal Form of a PlanePlane के perpendicular unit vector =  $\hat{n}$   
(given)Plane की origin से Distance =  $d$   
(given) $l, m, n \rightarrow D.C.$ Vector Form

$$\vec{r} \cdot \hat{n} = d$$

variable =  $x\hat{i} + y\hat{j} + z\hat{k}$ Cartesian Form

$$lx + my + nz = d$$

$$(l^2 + m^2 + n^2 = 1)$$

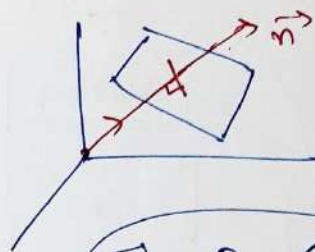
$$\vec{r} \cdot \hat{n} = d$$

e.g. Find the vector equation of a plane which is at a distance of  $\frac{6}{\sqrt{29}}$  from the origin and its normal vector from origin is  $2\hat{i} - 3\hat{j} + 4\hat{k}$ . Also find its Cartesian form.

$$d = \frac{6}{\sqrt{29}}$$

$$\vec{n} = 2\hat{i} - 3\hat{j} + 4\hat{k}$$

$$\hat{n} = \frac{2\hat{i} - 3\hat{j} + 4\hat{k}}{\sqrt{4+9+16}} = \frac{2\hat{i} - 3\hat{j} + 4\hat{k}}{\sqrt{29}}$$



$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

Vector form,  $\vec{r} \cdot \hat{n} = d$

$$\Rightarrow \vec{r} \cdot \left( \frac{2\hat{i} - 3\hat{j} + 4\hat{k}}{\sqrt{29}} \right) = \frac{6}{\sqrt{29}} \quad \checkmark$$

Cartesian form,

$$\frac{2x}{\sqrt{29}} - \frac{3y}{\sqrt{29}} + \frac{4z}{\sqrt{29}} = \frac{6}{\sqrt{29}} \quad \checkmark$$

e.g. Find the distance of the plane  $2x - 3y + 4z - 6 = 0$  from origin. Also find D.R. & D.C. of vector perpendicular to it.

$$\text{D.R.} = 2, -3, 4$$

$$2x - 3y + 4z = 6$$

$$\vec{r} \cdot (2\hat{i} - 3\hat{j} + 4\hat{k}) = 6$$

$$\Rightarrow \vec{r} \cdot \left( \frac{2\hat{i} - 3\hat{j} + 4\hat{k}}{\sqrt{29}} \right) = \frac{6}{\sqrt{29}}$$

$$\vec{r} \cdot \hat{n} = d$$

$$\text{Distance from origin} = \frac{6}{\sqrt{29}}$$

$$\frac{2x - 3y + 4z}{\sqrt{29}} = \frac{6}{\sqrt{29}}$$

$$\frac{2x}{\sqrt{29}} - \frac{3y}{\sqrt{29}} + \frac{4z}{\sqrt{29}} = \frac{6}{\sqrt{29}}$$

$$\text{D.C.} = \frac{2}{\sqrt{29}}, -\frac{3}{\sqrt{29}}, \frac{4}{\sqrt{29}}$$



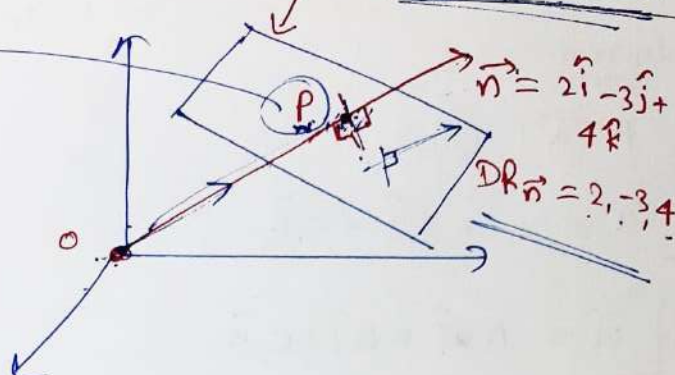
e.g. Find the coordinate of the foot of the perpendicular drawn ~~to~~ from the origin to the plane  $2x - 3y + 4z - 6 = 0$

Ans: Let  $P(x_1, y_1, z_1) = ?$

$O(0, 0, 0)$

$\vec{OP} \rightarrow$  Direction Ratios

$DR_{OP} = (x_1, y_1, z_1)$



Plane  $(2x - 3y + 4z - 6 = 0)$   $\propto$  Perpendicular vector  $(\vec{n})$   
 $\propto$  DR.  $= (2, -3, 4)$

$$\vec{OP} \parallel \vec{n} \rightarrow \begin{array}{c} DR_{OP} \propto DR_{\vec{n}} \\ \downarrow \qquad \qquad \downarrow \\ x_1, y_1, z_1 \qquad 2, -3, 4 \end{array}$$

$$\Rightarrow \frac{x_1}{2} = \frac{y_1}{-3} = \frac{z_1}{4} = \lambda$$

$$\boxed{x_1 = 2\lambda} \quad \boxed{y_1 = -3\lambda} \quad \boxed{z_1 = 4\lambda}$$

$\therefore$  Point 'P'  $(x_1, y_1, z_1)$  lies on the plane also.

$$2x - 3y + 4z - 6 = 0$$

$$\Rightarrow 2(x_1) - 3y_1 + 4z_1 - 6 = 0$$

$$\Rightarrow 2(2\lambda) - 3(-3\lambda) + 4(4\lambda) - 6 = 0$$

$$\Rightarrow 4\lambda + 9\lambda + 16\lambda = 6$$

$$\Rightarrow 29\lambda = 6$$

$$\lambda = \frac{6}{29}$$

$$P(x_1, y_1, z_1)$$

$$P(2\lambda, -3\lambda, 4\lambda)$$

$$P\left(\frac{12}{29}, -\frac{18}{29}, \frac{24}{29}\right)$$

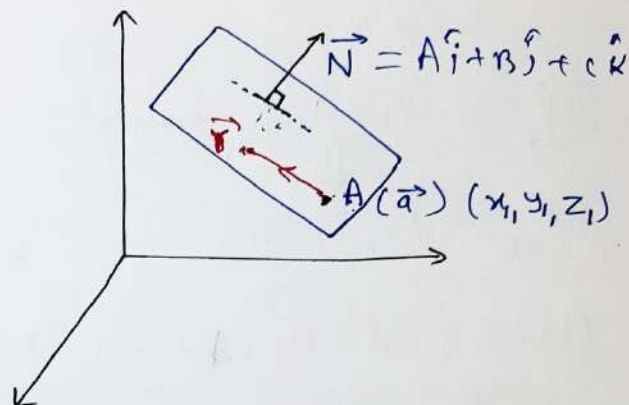
# Equation of a plane (Given - 1 Point, Direction)

(Plane passes through this) (Perpendicular to plane)

Given

$$\begin{aligned} &A(\vec{a}) (x_1, y_1, z_1) \\ &\vec{a} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k} \end{aligned}$$

$$\vec{N} = A\hat{i} + B\hat{j} + C\hat{k}$$



Plane:

Vector Form

$$(\vec{r} - \vec{a}) \cdot \vec{N} = 0$$

Point

Direction (Perpendicular)

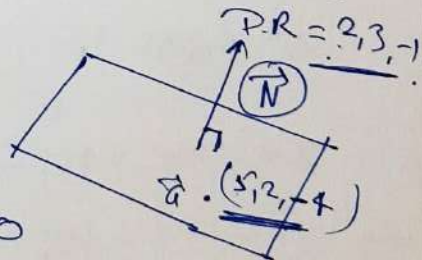
Cartesian Form

$$A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$$

E.g. Find the vector and Cartesian eq<sup>n</sup> of the plane which passes through the point  $(5, 2, -4)$  and perpendicular to the line with direction ratios 2, 3, -1.

Cartesian Form:

$$\begin{aligned} &2(x - 5) + 3(y - 2) - 1(z + 4) = 0 \\ \Rightarrow &2x + 3y - z = 20 \end{aligned}$$



Vector Form:  $(\vec{r} - \vec{a}) \cdot \vec{N} = 0$

$$\Rightarrow (\vec{r} - 5\hat{i} - 2\hat{j} + 4\hat{k}) \cdot (2\hat{i} + 3\hat{j} - \hat{k}) = 0$$



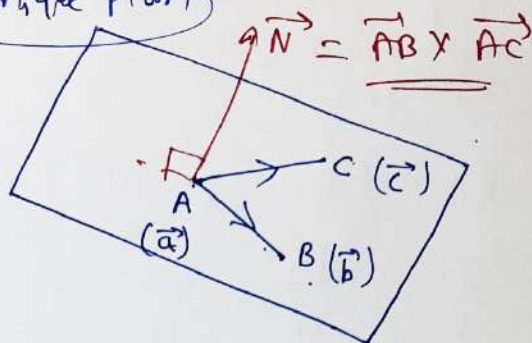
## Equation of a plane

Given

3 non-collinear ~~plane~~ points

| Points | Position Vector | Coordinates       |
|--------|-----------------|-------------------|
| A      | $\vec{a}$       | $(x_1, y_1, z_1)$ |
| B      | $\vec{b}$       | $(x_2, y_2, z_2)$ |
| C      | $\vec{c}$       | $(x_3, y_3, z_3)$ |

Unique Plane



## Plane

### Vector Form

$$(\vec{r} - \vec{a}) \cdot [(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})] = 0$$

### Cartesian Form

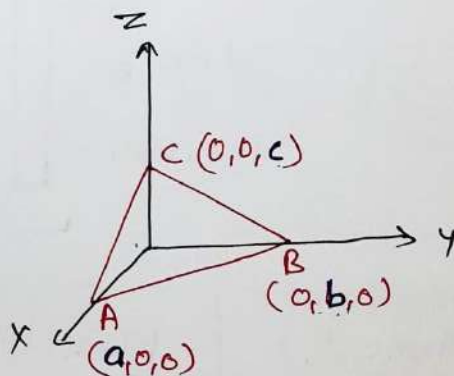
$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$$

## Intercept Form

x-intercept = a

y-intercept = b

z-intercept = c



## Equation of plane in Intercept form

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

e.g. Find  $x, y$  and  $z$ -intercepts of the plane

$$\vec{r} \cdot (6\hat{i} + 4\hat{j} + 3\hat{k}) = 12.$$

$$x\hat{i} + y\hat{j} + z\hat{k}$$

$$\Rightarrow 6x + 4y + 3z = 12$$

$$x\text{-intercept} = 2$$

$$\Rightarrow \frac{6x}{12} + \frac{4y}{12} + \frac{3z}{12} = 1$$

$$y \text{ ————— } = 3$$

$$z \text{ ————— } = 4$$

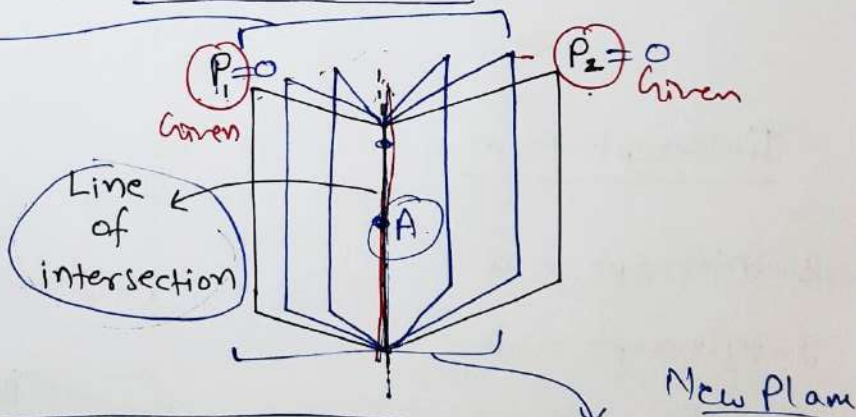
$$\Rightarrow \boxed{\frac{x}{2} + \frac{y}{3} + \frac{z}{4} = 1}$$

Family of Planes

(New plane passing through the intersection of two given planes)

New Plane

$$\boxed{P_1 + \lambda P_2 = 0}$$



|                | Plane ( $P_1$ )                 | Plane ( $P_2$ )                 | Family of planes ( $P_1 + \lambda P_2 = 0$ )                                   |
|----------------|---------------------------------|---------------------------------|--------------------------------------------------------------------------------|
| Vector Form    | $\vec{r} \cdot \vec{n}_1 = d_1$ | $\vec{r} \cdot \vec{n}_2 = d_2$ | $(\vec{r} \cdot \vec{n}_1 - d_1) + \lambda(\vec{r} \cdot \vec{n}_2 - d_2) = 0$ |
| Cartesian Form | $A_1x + B_1y + C_1z = d_1$      | $A_2x + B_2y + C_2z = d_2$      | $(A_1x + B_1y + C_1z - d_1) + \lambda(A_2x + B_2y + C_2z - d_2) = 0$           |



e.g. Find the Cartesian equation of the plane New passing through the intersection of the planes  $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 6$  and  $\vec{r} \cdot (2\hat{i} + 3\hat{j} + 4\hat{k}) = -5$ , and also passes through origin.

New plane  $(P_3) = 0$

$$\Rightarrow P_1 + \lambda(P_2) = 0$$

$$\Rightarrow (x+y+z-6) + \lambda(2x+3y+4z+5) = 0$$

$(0,0,0)$  satisfies —

$$\Rightarrow (0-6) + \lambda(0+5) = 0$$

$$\Rightarrow \boxed{\lambda = \frac{6}{5}}$$

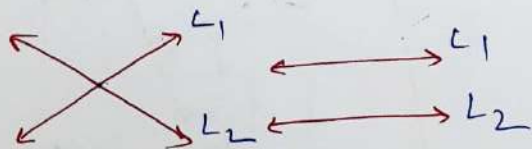
$$\begin{cases} P_1 \equiv x+y+z-6=0 \\ P_2 \equiv 2x+3y+4z+5=0 \end{cases}$$

New plane

$$(x+y+z-6) + \frac{6}{5}(2x+3y+4z) = 0$$

## Coplanarity of two lines

### Coplanar Lines



Yes, Point of intersection

Parallel

$$L_1: \vec{r} = \vec{a}_1 + \lambda \vec{b}_1$$

$$L_2: \vec{r} = \vec{a}_2 + \mu \vec{b}_2$$

$$d=0$$

Point

Vector Form

direction

Condition of coplanarity of two lines

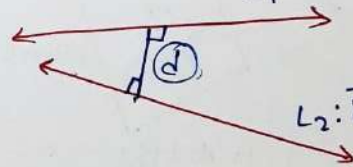
$$\boxed{(a_2 - a_1) \cdot (b_1 \times b_2) = 0}$$

### Non-Coplanar

Skew lines

$$L_1: \vec{r} = \vec{a}_1 + \lambda \vec{b}_1$$

$$L_2: \vec{r} = \vec{a}_2 + \mu \vec{b}_2$$



- No point of intersection
- not parallel

$$d = \left| \frac{(a_2 - a_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|} \right| \neq 0$$

$$d \neq 0$$

e.g. Show that the lines  $\frac{x+3}{-3} = \frac{y-1}{1} = \frac{z-5}{5}$  and

$\frac{x+1}{-1} = \frac{y-2}{2} = \frac{z-5}{5}$  are coplanar.  $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0$

$$\vec{b}_1 = -3\hat{i} + \hat{j} + 5\hat{k}$$

$$\vec{b}_2 = -\hat{i} + 2\hat{j} + 5\hat{k}$$

$$\vec{a}_1 = -3\hat{i} + \hat{j} + 5\hat{k}$$

$$\vec{a}_2 = -\hat{i} + 2\hat{j} + 5\hat{k}$$

$$\vec{a}_2 - \vec{a}_1 = 2\hat{i} + \hat{j} + 0\hat{k}$$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix}$$

$$\vec{b}_1 \times \vec{b}_2 = -5\hat{i} + 10\hat{j} - 5\hat{k}$$

Cond<sup>n</sup>. of Coplanarity.

$$(2\hat{i} + \hat{j}) \cdot (-5\hat{i} + 10\hat{j} - 5\hat{k}) = -10 + 10 = 0$$

Distance of a Point from a Plane.

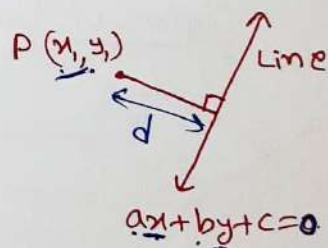
(3D)



$$Ax + By + Cz = D$$

$$d = \left| \frac{Ax_1 + By_1 + Cz_1 - D}{\sqrt{A^2 + B^2 + C^2}} \right|$$

(2D)



$$ax + by + c = 0$$

$$d = \left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$$

e.g. Find the distance b/w Point  $(2, 5, -3)$  and plane

$$\vec{r} \cdot (6\hat{i} - 3\hat{j} + 2\hat{k}) = 4$$

$$d = \left| \frac{12 - 15 - 6 - 4}{\sqrt{6^2 + (-3)^2 + 2^2}} \right|$$

$$= \left| \frac{-13}{7} \right| = \frac{13}{7}$$

$$6x - 3y + 2z - 4 = 0$$



Angle between   &  

Dot Product  
 $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$

|       |       |               |
|-------|-------|---------------|
| Line  | Line  | $\cos \theta$ |
| Plane | Plane | $\cos \theta$ |
| Plane | Line  | <u>SMD</u> ?  |

Already Covered

I Angle between Two Lines

$\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$   
 $\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$  (L1)  
 $\vec{r} = \vec{a}_2 + \mu \vec{b}_2$   
 $\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$  (L2)

$$\cos \theta = \left| \frac{\vec{b}_1 \cdot \vec{b}_2}{|\vec{b}_1| |\vec{b}_2|} \right|$$

$$\cos \theta = \left| \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}} \right|$$

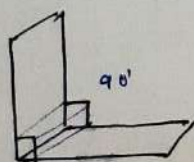
II Angle between Two Planes

$\vec{r} \cdot \vec{n}_1 = d_1$   
 $A_1 x + B_1 y + C_1 z + D_1 = 0$   
 $\vec{r} \cdot \vec{n}_2 = d_2$   
 $A_2 x + B_2 y + C_2 z + D_2 = 0$

$$\cos \theta = \left| \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1| |\vec{n}_2|} \right|$$

$$\cos \theta = \left| \frac{A_1 A_2 + B_1 B_2 + C_1 C_2}{\sqrt{A_1^2 + B_1^2 + C_1^2} \sqrt{A_2^2 + B_2^2 + C_2^2}} \right|$$

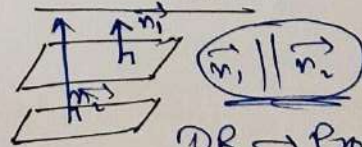
Note: ~~Per~~ Perpendicular Planes



$\theta = 90^\circ$   
 $\cos \theta = 0$

$$A_1 A_2 + B_1 B_2 + C_1 C_2 = 0$$

Parallel planes



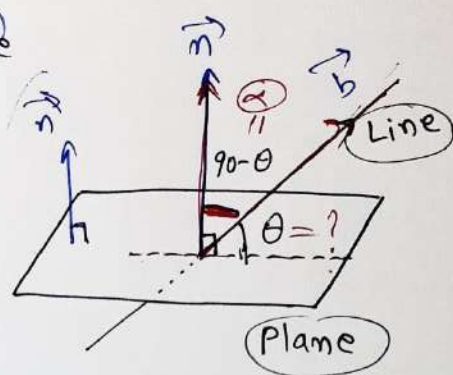
DR  $\rightarrow$  Proportional

$$\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2}$$

### III Angle between a line & a plane

Line  $\vec{r} = \vec{a} + \lambda \vec{b}$

Plane,  $\vec{r} \cdot \vec{n} = d$



$$\cos(90 - \theta) = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}| |\vec{n}|}$$

$$\Rightarrow \sin \theta = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}| |\vec{n}|}$$

e.g. Find the angle between the planes  $2x + y - 2z = 5$  and  $3x - 6y - 2z = 7$ . ( $P_1$ ) ( $P_2$ )

$$\cos \theta = \frac{|(2 \times 3) + 1 \times (-6) + (-2) \times (-2)|}{\sqrt{2^2 + 1^2 + (-2)^2} \sqrt{3^2 + (-6)^2 + (-2)^2}} = \frac{|6 - 6 + 4|}{\sqrt{9} \sqrt{49}}$$

$$\cos \theta = \frac{4}{21} \Rightarrow \theta = \cos^{-1}\left(\frac{4}{21}\right)$$

e.g. Find the angle between the line  $\frac{x+1}{2} = \frac{y}{3} = \frac{z-3}{6}$  and the plane  $10x + 2y - 11z = 3$ .

Line direction, D.R. 2, 3, 6 ( $\vec{b} = 2\hat{i} + 3\hat{j} + 6\hat{k}$ )

Plane direction, D.R. 10, 2, -11 ( $\vec{n} = 10\hat{i} + 2\hat{j} - 11\hat{k}$ )

$$\sin \theta = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}| |\vec{n}|} = \frac{(2\hat{i} + 3\hat{j} + 6\hat{k}) \cdot (10\hat{i} + 2\hat{j} - 11\hat{k})}{\sqrt{2^2 + 3^2 + 6^2} \sqrt{10^2 + 2^2 + (-11)^2}}$$

$$\sin \theta = \frac{|20 + 6 - 66|}{7 \times 15} \Rightarrow \sin \theta = \frac{40}{7 \times 15} \Rightarrow \theta = \sin^{-1}\left(\frac{8}{21}\right)$$

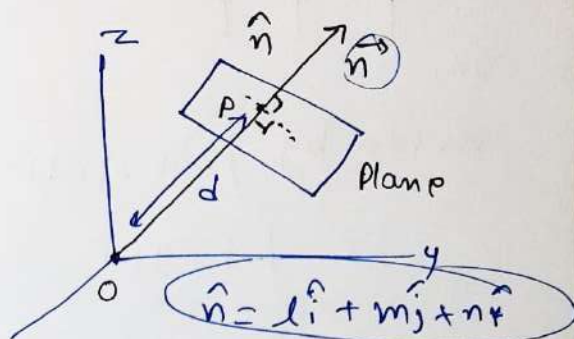


## Exercise 11.3

Plane (3D)Eq<sup>n</sup> of plane,

$$\vec{r} \cdot \hat{n} = d$$

Vector Form

Direction  
✓Distance  
from originUnit vector  
Perpendicular  
to the plane

$$l^2 + m^2 + n^2 = 1$$

$$lx + my + nz = d$$

Cartesian Form.

D.C.

Distance from  
origin

e.g.

$$Ax + By + Cz = d$$

D.R.

A, B, C

$$\left( \frac{A}{\sqrt{A^2 + B^2 + C^2}}, \frac{B}{\sqrt{A^2 + B^2 + C^2}}, \frac{C}{\sqrt{A^2 + B^2 + C^2}} \right)$$

## Exercise 11.3

Q.1 In each of the following cases, determine the direction cosines of the normal to the plane and the distance from the origin.

(a)  $z = 2$

$$\Rightarrow 0x + 0y + 1z = 2$$

$$\Rightarrow \text{D.C. } [l=0, m=0, n=1]$$

$$l^2 + m^2 + n^2 = 1$$

Distance from  
origin = 2

(b)  $x + y + z = 1$

D.R. 1, 1, 1

Divide by  $\sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$   
on both sides,

$$\Rightarrow \frac{x}{\sqrt{3}} + \frac{y}{\sqrt{3}} + \frac{z}{\sqrt{3}} = \frac{1}{\sqrt{3}}$$

$$\text{D.C.} = \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$$

Distance from (0,0,0)  
=  $\frac{1}{\sqrt{3}}$

$$\textcircled{C} \quad 2x + 3y - z = 5$$

DR.

Divide by  $\sqrt{2^2 + 3^2 + (-1)^2}$

$$= \sqrt{4 + 9 + 1}$$

$$= \sqrt{14} \text{ in LHS \& RHS}$$

$$\text{Eq.} \quad \left[ \frac{2x}{\sqrt{14}} + \frac{3y}{\sqrt{14}} - \frac{z}{\sqrt{14}} = \frac{5}{\sqrt{14}} \right]$$

D.C.  $\frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}, -\frac{1}{\sqrt{14}}$

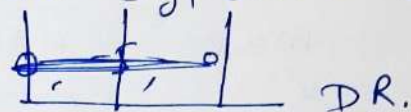
$$d = \frac{5}{\sqrt{14}}$$

$$\textcircled{D} \quad 5y + 8 = 0$$

$$\Rightarrow 5y = -8$$

$$\Rightarrow -5y = 8$$

$$\Rightarrow 0.x - 5.y + 0.z = 8$$



DR.

Divide by  $\sqrt{0^2 + (-5)^2 + 0^2}$

$$= \sqrt{25} = 5$$

$$\Rightarrow \frac{0.x}{5} - \frac{5y}{5} + \frac{0.z}{5} = \frac{8}{5}$$

D.C.  $0, -1, 0$

$$d = \frac{8}{5}$$

Q.2 Find the vector equation of a plane which is at a distance 7 units from the origin and normal to the vector  $3\hat{i} + 5\hat{j} - 6\hat{k}$ .

$$\textcircled{d=7}$$

$$\vec{r} \cdot \hat{n} = d$$

unit vector

$$\Rightarrow \left[ \vec{r} \cdot \left( \frac{3\hat{i} + 5\hat{j} - 6\hat{k}}{\sqrt{70}} \right) = 7 \right]$$

normal vector ( $\vec{n}$ )

$$\hat{n} = \frac{\vec{n}}{|\vec{n}|} = \frac{3\hat{i} + 5\hat{j} - 6\hat{k}}{\sqrt{9 + 25 + 36}}$$

$$= \frac{3\hat{i} + 5\hat{j} - 6\hat{k}}{\sqrt{70}}$$



**Q.3** Find the Cartesian ~~Point~~ equation of the following planes:

(a)  $\vec{r} \cdot (\hat{i} + \hat{j} - \hat{k}) = 2$  Vector Form.

$(\vec{r} = x\hat{i} + y\hat{j} + z\hat{k})$  put.

$$\Rightarrow (x\hat{i} + y\hat{j} + z\hat{k}) \cdot (\hat{i} + \hat{j} - \hat{k}) = 2$$

$$\Rightarrow \boxed{x + y - z = 2}$$

Cartesian Form,

(b)  $\vec{r} \cdot (2\hat{i} + 3\hat{j} - 4\hat{k}) = 1$

Cartesian Form,

$$\boxed{2x + 3y - 4z = 1}$$

(c)  $\vec{r} \cdot [(s-2t)\hat{i} + (3-t)\hat{j} + (2s+t)\hat{k}] = 5$

Vector Form

Cartesian form,

$$\boxed{(s-2t)x + (3-t)y + (2s+t)z = 5}$$

~~**Q.4** In the following cases, find the coordinates~~

## Exercise 11.3

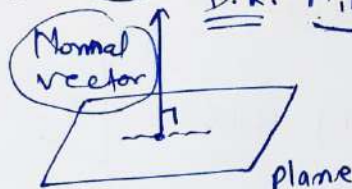
## Plane (3D)

Plane

$$[\vec{r}, \vec{n}] = d$$

Vector form

$$A\hat{i} + B\hat{j} + C\hat{k} = \vec{n}$$

D.R.  $A, B, C$ 

$$Ax + By + Cz = d$$

Cartesian form

Q.4 In the following cases, find the coordinates of the foot of perpendicular drawn from the origin.

(i)  $2x + 3y + 4z - 12 = 0$

Foot of perpendicular drawn from origin =  $P(x_1, y_1, z_1)$

(let)

$$\vec{n} = 2\hat{i} + 3\hat{j} + 4\hat{k}$$

$$DR_{\vec{n}} = 2, 3, 4$$

$$DR_{\vec{OP}} = x_1, y_1, z_1$$

$$\vec{n} \parallel \vec{OP} \rightarrow DR \rightarrow (\text{Proportional})$$

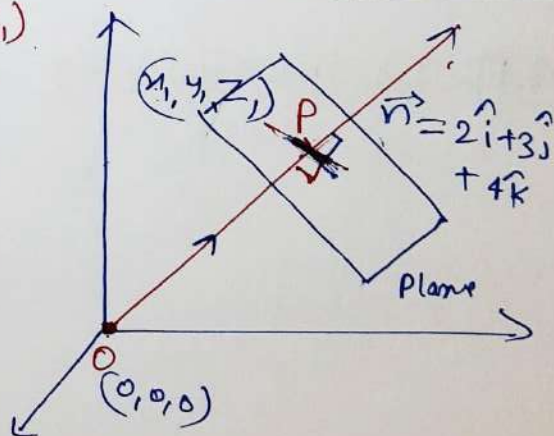
$$P(x_1, y_1, z_1) ?$$

$$P(2\lambda, 3\lambda, 4\lambda)$$

$\uparrow \quad \uparrow \quad \uparrow$

$$A(x_1, y_1, z_1) \rightarrow B(x_2, y_2, z_2)$$

$$DR_{AB} = x_2 - x_1, y_2 - y_1, z_2 - z_1$$



$$\frac{x_1}{2} = \frac{y_1}{3} = \frac{z_1}{4} = \lambda \quad (\text{let})$$

$$\underline{x_1 = 2\lambda}, \underline{y_1 = 3\lambda}, \underline{z_1 = 4\lambda}$$



Plane  $2x + 3y + 4z - 12 = 0$

Point 'P' lies on plane also.

Foot of Perpendicular

$P(x_1, y_1, z_1)$

or

$P(2\lambda, 3\lambda, 4\lambda)$

$\Rightarrow$  Point 'P'  $(2\lambda, 3\lambda, 4\lambda)$  will satisfy the eq<sup>n</sup> of plane  $\boxed{2x + 3y + 4z - 12 = 0}$

$\Rightarrow 2(2\lambda) + 3(3\lambda) + 4(4\lambda) - 12 = 0$

$\Rightarrow 4\lambda + 9\lambda + 16\lambda = 12 \Rightarrow 29\lambda = 12 \Rightarrow \lambda = \frac{12}{29}$

Foot of Perpendicular drawn from origin

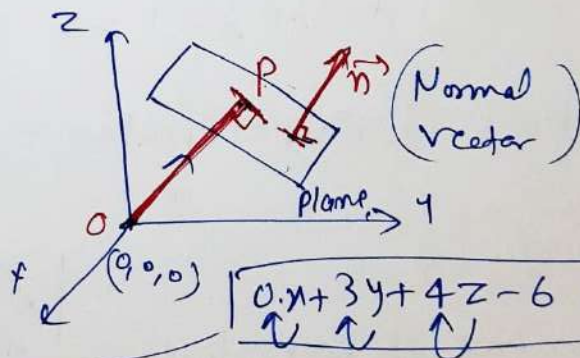
$P(x_1, y_1, z_1) \equiv P(2\lambda, 3\lambda, 4\lambda)$

$\equiv P\left(\frac{24}{29}, \frac{36}{29}, \frac{48}{29}\right)$

(b)  $\boxed{3y + 4z - 6 = 0}$

Foot of Perpendicular

$P(x_1, y_1, z_1)$



$\vec{OR}_{OP} = x_1, y_1, z_1$

$\vec{OR}_{\vec{n}} = 0, 3, 4$

$\therefore \vec{OP} \parallel \vec{n} \rightarrow \vec{OR} \rightarrow \text{Proportional} \quad \frac{x_1}{0} = \frac{y_1}{3} = \frac{z_1}{4} = \lambda$  (let)

$P(0, 3\lambda, 4\lambda)$  satisfy the eq<sup>n</sup> of plane,  $3y + 4z - 6 = 0$

$\Rightarrow 3(3\lambda) + 4(4\lambda) - 6 = 0 \Rightarrow 25\lambda = 6 \Rightarrow \lambda = \frac{6}{25}$

Foot of perpendicular  $P(0, 3\lambda, 4\lambda)$

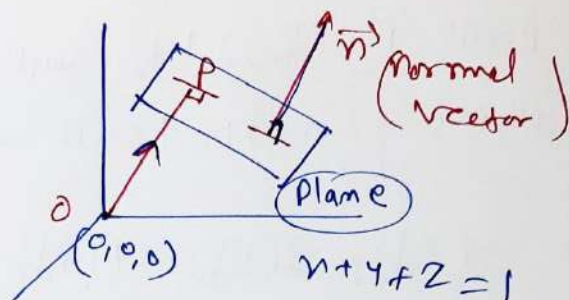
$$\lambda = \frac{6}{25}$$

$$\therefore P(0, \frac{18}{25}, \frac{24}{25})$$

(C)  $x+y+z=1$

Foot of perpendicular

$$P(x_1, y_1, z_1)$$



$$DR \vec{OP} = x_1, y_1, z_1$$

$$DR \vec{n} = 1, 1, 1$$

$$\vec{OP} \parallel \vec{n} \rightarrow DR = \text{Proportional} \rightarrow \frac{x_1}{1} = \frac{y_1}{1} = \frac{z_1}{1} = \lambda \text{ (let)}$$

$$P(x_1, y_1, z_1) = P(\lambda, \lambda, \lambda)$$

Point  $P(\lambda, \lambda, \lambda)$  lies on  $(x+y+z=1)$  plane.

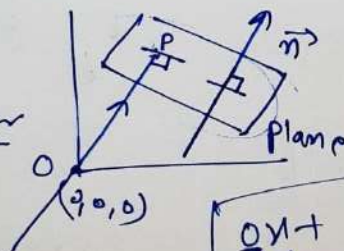
$$\Rightarrow \lambda + \lambda + \lambda = 1 \Rightarrow 3\lambda = 1 \Rightarrow \lambda = \frac{1}{3}$$

$$\text{Foot of perpendicular} = P(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$$

(d)  $5y+8=0$

$P \rightarrow$  foot of perpendicular

$$(x_1, y_1, z_1)$$



$$DR \vec{OP} = x_1, y_1, z_1$$

$$DR \vec{n} = 0, 5, 0$$

$$\vec{OP} \parallel \vec{n} \rightarrow DR \rightarrow \text{Proportional} \rightarrow \frac{x_1}{0} = \frac{y_1}{5} = \frac{z_1}{0} = \lambda$$

$$x_1 = 0, y_1 = 5\lambda, z_1 = 0$$



Foot of Perpendicular  $P(x, y, z)$   
or  $P(0, 5\lambda, 0)$

$$5\lambda = 8 \times \frac{-8}{2 \times 5}$$

Plane  $5y + 8 = 0$

Satisfy,

$$P\left(0, -\frac{8}{5}, 0\right)$$

$$\Rightarrow 5(5\lambda) + 8 = 0$$

$$\Rightarrow 25\lambda = -8$$

$$\Rightarrow \lambda = -\frac{8}{25}$$

Q.5 Find the Vector & Cartesian eq<sup>n</sup>. of the  
Planes

(a) that passes through the point  $(1, 0, -2)$  and  
the normal to the plane is  $\hat{i} + \hat{j} - \hat{k}$ .

Standard form,

normal vector  $(\vec{n})$

$$\vec{n} = A\hat{i} + B\hat{j} + C\hat{k}$$

$$D.R.(A, B, C)$$

Cartesian form,

$$A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$$

Vector form  
 $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$

$$P_0(\vec{a})$$

$$(x_1, y_1, z_1)$$

Point  $(1, 0, -2)$

$$\vec{a} = \hat{i} - 2\hat{k}$$

$$\vec{n} = \hat{i} + \hat{j} - \hat{k} \quad (D.R. \Rightarrow 1, 1, -1)$$

Vector Form,

$$(\vec{r} - \vec{a}) \cdot \vec{n} = 0$$

$$[\vec{r} - (\hat{i} - 2\hat{k})] \cdot (\hat{i} + \hat{j} - \hat{k}) = 0$$

Cartesian form,

$$1(x - 1) + 1(y - 0) - 1(z + 2) = 0$$

$$\Rightarrow x - 1 + y - z - 2 = 0$$

$$\Rightarrow x + y - z = 3$$

(b) that passes through the point  $(1, 4, 6)$  and the normal vector to the plane is  $\hat{i} - 2\hat{j} + \hat{k}$ .

Point  $(1, 4, 6)$   
 $\vec{a} = \hat{i} + 4\hat{j} + 6\hat{k}$

Normal vector

$$\vec{n} = \hat{i} - 2\hat{j} + \hat{k}$$

(D.R. =  $\underset{A}{1}, \underset{B}{-2}, \underset{C}{1}$ )

Vector Form:

$$(\vec{r} - \vec{a}) \cdot \vec{n} = 0$$

$$\Rightarrow [\vec{r} - (\hat{i} + 4\hat{j} + 6\hat{k})] \cdot (\hat{i} - 2\hat{j} + \hat{k}) = 0$$

Cartesian Form

$$\textcircled{A}(x-x_1) + \textcircled{B}(y-y_1) + \textcircled{C}(z-z_1) = 0$$

$$\Rightarrow 1(x-1) + (-2)(y-4) + 1(z-6) = 0$$

$$\Rightarrow (x-1) - 2y + 8 + z - 6 = 0$$

$$\Rightarrow x - 2y + z - 1 = 0$$

**Q.6** Find the equations of the planes that passes through three points.

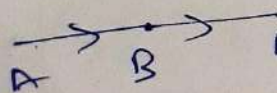
(a)  $\underset{A}{(1, 1, -1)}, \underset{B}{(6, 4, -5)}, \underset{C}{(-4, -2, 3)} \rightarrow$



$$\left. \begin{array}{l} DR_{AB} = 5, 3, -4 \\ DR_{BC} = -10, -6, 8 \end{array} \right\} \frac{5}{-10} = \frac{3}{-6} = \frac{-4}{8}$$

DR  $\rightarrow$  Proportional

$$-\frac{1}{2} = -\frac{1}{2} = -\frac{1}{2}$$



$\rightarrow$  Collinear  $\rightarrow$

$\infty$  no. of planes are possible



(b)  $\underset{A}{(1,1,0)}, \underset{B}{(1,2,1)}, \underset{C}{(-2,2,-1)}$

$\Rightarrow R_{AB} = 0, 1, -1$   $\left\{ \begin{array}{l} \frac{0}{-3} \neq \frac{1}{0} \neq \frac{-1}{-2} \end{array} \right.$

$\Rightarrow R_{BC} = -3, 0, -2$

$A, B, C \rightarrow$  non-collinear points.

Unique Plane  $\checkmark$

Eq<sup>n</sup>. of plane

$A (1,1,0)$   
 $B (1,2,1)$   
 $C (-2,2,-1)$

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x-1 & y-1 & z-0 \\ 1-1 & 2-1 & 1-0 \\ -2-1 & 2-1 & -1-0 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x-1 & y-1 & z \\ 0 & 1 & 1 \\ -3 & 1 & -1 \end{vmatrix} = 0$$

$$\Rightarrow (x-1)[-1-1] - (y-1)(3) + z(3) = 0$$

$$\Rightarrow (x-1)(-2) - 3y + 3 + 3z = 0$$

$$\Rightarrow -2x + 2 - 3y + 3 + 3z = 0$$

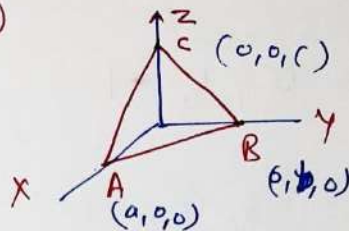
$$\Rightarrow \boxed{5 = 2x + 3y - 3z}$$

## Exercise 11.3

## Plane (3D)

Intercept Form.

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$



Q.7 Find the intercepts cut off by the plane  
 $2x + y - z = 5$ .

$$\Rightarrow \frac{2x}{5} + \frac{y}{5} - \frac{z}{5} = 1$$

$$\Rightarrow \left[ \frac{x}{(5/2)} + \frac{y}{(5)} + \frac{z}{(-5)} = 1 \right]$$

$$x\text{-intercept} = \frac{5}{2} \checkmark$$

$$y\text{-intercept} = 5 \checkmark$$

$$z\text{-intercept} = -5 \checkmark$$

Q.8 Find the equation of the plane with intercept  
 3 on y-axis and parallel to ZOx plane.

$$y\text{-intercept} = 3 = b$$

Intercept Form.

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

$$\Rightarrow \frac{x}{\infty} + \frac{y}{3} + \frac{z}{\infty} = 1$$

x-intercept

 $\infty$ 

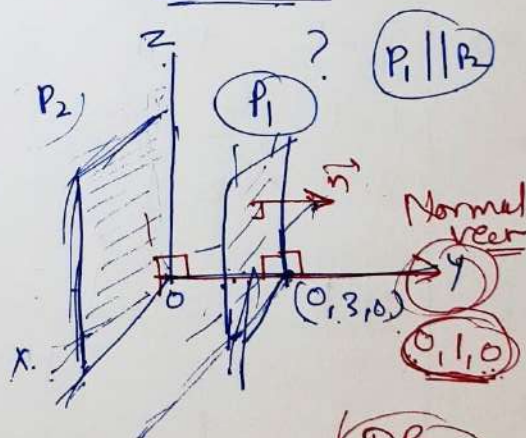
z-intercept

 $\infty$ 

$$\Rightarrow \frac{x}{\infty} + \frac{y}{3} + \frac{z}{\infty} = 1$$

$$\Rightarrow 0 + \frac{y}{3} + 0 = 1$$

$$\Rightarrow y = 3$$



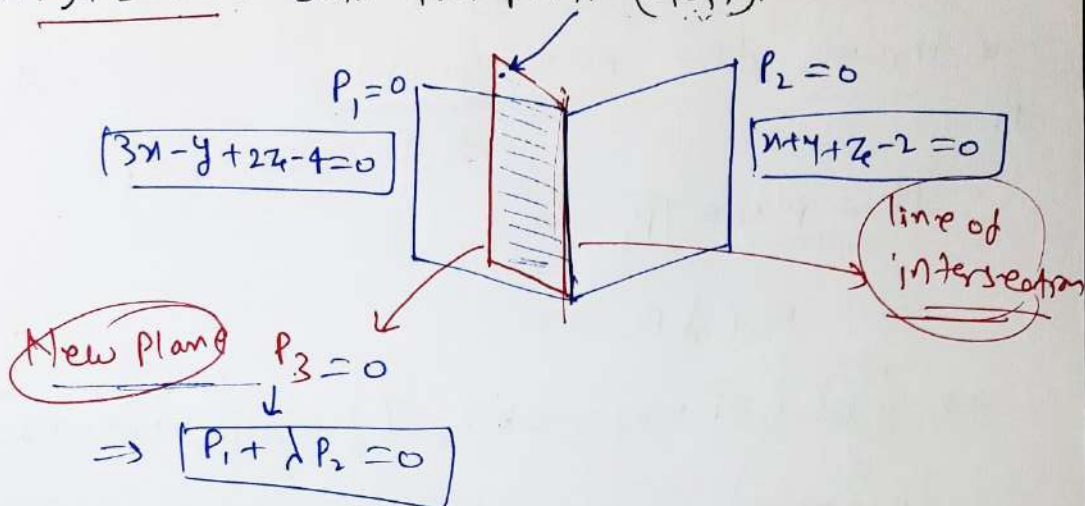
DR

$$\frac{\frac{1}{a}, \frac{1}{b}, \frac{1}{c}}{0, 1, 0}$$

$$\frac{\frac{1}{a}}{0} = \frac{\frac{1}{b}}{1} = \frac{\frac{1}{c}}{0}$$



Q.9 Find the equation of the plane passing through the intersection of the planes  $3x - y + 2z - 4 = 0$  and  $x + y + z - 2 = 0$  and the point  $(2, 2, 1)$ .



$$\Rightarrow (3x - y + 2z - 4) + \lambda(x + y + z - 2) = 0$$

Also this plane passes through  $(2, 2, 1)$

$$\Rightarrow (6 - 2 + 2 - 4) + \lambda(2 + 2 + 1 - 2) = 0$$

$$\Rightarrow 2 + \lambda(3) = 0$$

$$\Rightarrow \lambda = -\frac{2}{3}$$

Required plane:  $(3x - y + 2z - 4) - \frac{2}{3}(x + y + z - 2) = 0$

$$\Rightarrow \frac{9x - 3y + 6z - 12 - 2x - 2y - 2z + 4}{3} = 0$$

$$\Rightarrow 7x - 5y + 4z - 8 = 0$$

Q. 10 Find the vector equation of the plane passing through the intersection of the planes

$$\vec{r} \cdot (2\hat{i} + 2\hat{j} - 3\hat{k}) = 7 \quad \& \quad \vec{r} \cdot (2\hat{i} + 5\hat{j} + 3\hat{k}) = 9 \quad \text{and}$$

through the point  $(2, 1, 3)$ .

$P_1$

$P_2$

$$\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k}$$

New plane  $P_3 = 0$

$$\Rightarrow P_1 + \lambda P_2 = 0$$

$$\Rightarrow \underbrace{[\vec{r} \cdot (2\hat{i} + 2\hat{j} - 3\hat{k}) - 7]}_{\text{variable}} + \lambda \underbrace{[\vec{r} \cdot (2\hat{i} + 5\hat{j} + 3\hat{k}) - 9]}_{\text{variable}} = 0$$

(this plane passes through the point

$$\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k})$$

$$\Rightarrow (2\hat{i} + \hat{j} + 3\hat{k}) \cdot (2\hat{i} + 2\hat{j} - 3\hat{k}) - 7 + \lambda [(2\hat{i} + \hat{j} + 3\hat{k}) \cdot (2\hat{i} + 5\hat{j} + 3\hat{k}) - 9] = 0$$

$$\Rightarrow 4 + 2 - 9 - 7 + \lambda [4 + 5 + 9 - 9] = 0$$

$$\Rightarrow \lambda(9) = 10$$

$$\Rightarrow \lambda = \frac{10}{9}$$

Required Plane  $[\vec{r} \cdot (2\hat{i} + 2\hat{j} - 3\hat{k}) - 7]$

$$+ \left(\frac{10}{9}\right) [\vec{r} \cdot (2\hat{i} + 5\hat{j} + 3\hat{k}) - 9] = 0$$



$$\Rightarrow \frac{9[\vec{r} \cdot (2\hat{i} + 2\hat{j} - 3\hat{k}) - 7] + 10[\vec{r} \cdot (2\hat{i} + 5\hat{j} + 3\hat{k}) - 9]}{9} = 0$$

$$\Rightarrow \vec{r} \cdot (18\hat{i} + 18\hat{j} - 27\hat{k} + 20\hat{i} + 50\hat{j} + 30\hat{k})$$

$$-63 - 90 = 0$$

$$\Rightarrow \vec{r} \cdot (38\hat{i} + 68\hat{j} + 3\hat{k}) = 153$$

[Q.11] Find the eq<sup>n</sup>. of the plane through the line of intersection of the planes  $P_1: x+y+z=1$  and  $P_2: 2x+3y+4z=5$  which is perpendicular to the plane  $x-y+z=0$ .

Ans:  $P_1: x+y+z-1=0$

$P_2: 2x+3y+4z-5=0$

New plane  $P_3=0$

$$\Rightarrow P_1 + \lambda P_2 = 0$$

$$\Rightarrow (x+y+z-1) + \lambda(2x+3y+4z-5) = 0$$

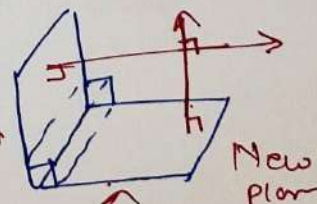
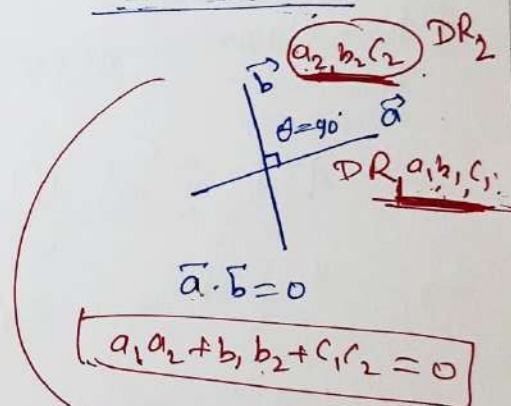
$$\Rightarrow x(1+2\lambda) + y(1+3\lambda) + z(1+4\lambda) - 1 - 5\lambda = 0$$

New plane

DR<sub>1</sub>:  $1+2\lambda, 1+3\lambda, 1+4\lambda$

$x-y+z=0$

DR<sub>2</sub>:  $1, -1, 1$



New Plane  $\vec{DR} = \underline{1+2\lambda, 1+3\lambda, 1+4\lambda}$

$(x-y+z)_{=0}$   $\vec{DR} = \underline{1, -1, 1}$

Condition of Perpendicular Planes

$$a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$$

$$\Rightarrow (1+2\lambda) - (1+3\lambda) + (1+4\lambda) = 0$$

$$\Rightarrow \cancel{1} + 2\lambda - \cancel{1} - 3\lambda + 1 + 4\lambda = 0$$

$$\Rightarrow 3\lambda + 1 = 0$$

$$\Rightarrow \lambda = -\frac{1}{3}$$

New plane  $x(1+2\lambda) + y(1+3\lambda) + z(1+4\lambda) - 1 - 5\lambda = 0$

$$\Rightarrow x\left(1 - \frac{2}{3}\right) + y\left(1 - \frac{3}{3}\right) + z\left(1 - \frac{4}{3}\right) - 1 + \frac{5}{3} = 0$$

$$\Rightarrow x \cdot \frac{1}{3} + 0 + z\left(-\frac{1}{3}\right) + \frac{2}{3} = 0$$

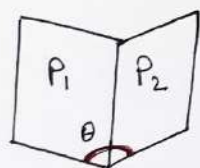
$$\Rightarrow \frac{x}{3} - \frac{z}{3} + \frac{2}{3} = 0$$

$$\Rightarrow \boxed{x - z + 2 = 0}$$



## Exercise 11.3

## Plane (3D)



## Angle b/w Two Planes

## Vector Form

$$\begin{cases} P_1: \vec{r} \cdot \vec{n}_1 = d_1 \\ P_2: \vec{r} \cdot \vec{n}_2 = d_2 \end{cases}$$

$$\cos \theta = \left| \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1| |\vec{n}_2|} \right|$$

## Cartesian Form

$$P_1: A_1x + B_1y + C_1z = d_1$$

$$P_2: A_2x + B_2y + C_2z = d_2$$

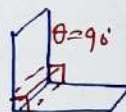
$$\cos \theta = \left| \frac{A_1A_2 + B_1B_2 + C_1C_2}{\sqrt{A_1^2 + B_1^2 + C_1^2} \cdot \sqrt{A_2^2 + B_2^2 + C_2^2}} \right|$$

## Condition for Parallel planes



$$\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2}$$

## Condition for Perpendicular Planes



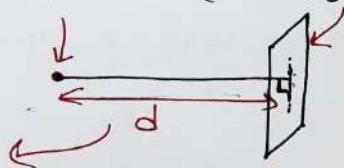
$$A_1A_2 + B_1B_2 + C_1C_2 = 0$$

## Distance Formula [b/w Point &amp; Plane]

(x<sub>1</sub>, y<sub>1</sub>, z<sub>1</sub>)

(Ax + By + Cz = d)

$$d = \left| \frac{Ax_1 + By_1 + Cz_1 - d}{\sqrt{A^2 + B^2 + C^2}} \right|$$



Q.12 Find the angle b/w the planes whose vector equations are

$$\vec{r} \cdot \underbrace{(2\hat{i} + 2\hat{j} - 3\hat{k})}_{\vec{n}_1} = 5 \quad \text{and} \quad \vec{r} \cdot \underbrace{(3\hat{i} - 3\hat{j} + 5\hat{k})}_{\vec{n}_2} = 3.$$

$$\cos \theta = \left| \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1| |\vec{n}_2|} \right| = \left| \frac{(2\hat{i} + 2\hat{j} - 3\hat{k}) \cdot (3\hat{i} - 3\hat{j} + 5\hat{k})}{\sqrt{4+4+9} \cdot \sqrt{9+9+25}} \right|$$

$$\Rightarrow \cos \theta = \left| \frac{6 - 6 - 15}{\sqrt{17} \sqrt{43}} \right| \Rightarrow \cos \theta = \frac{15}{\sqrt{731}} \Rightarrow \theta = \cos^{-1} \left( \frac{15}{\sqrt{731}} \right)$$

**Q.13** In the following cases, determine whether the given planes are parallel or perpendicular, and in case they are neither, find angles b/w them.

(a)  $\frac{7x+5y+6z+30=0}{x}$  &  $\frac{3x-y-10z+4=0}{x}$

Cond<sup>n</sup>. for Parallel Planes  $\frac{7}{3} \neq \frac{5}{-1} \neq \frac{6}{-10}$  Not Parallel -

Cond<sup>n</sup>. for Perpendicular Planes  $21 - 5 - 60 \neq 0$

$\Rightarrow 21 - 65 \neq 0$   
 (Not  $\perp$  planes)

$$\cos \theta = \left| \frac{21 - 5 - 60}{\sqrt{49 + 25 + 36} \sqrt{9 + 1 + 100}} \right| = \left| \frac{44}{\sqrt{110} \sqrt{110}} \right|$$

$$\cos \theta = \frac{44}{110} \Rightarrow \cos \theta = \frac{2}{5} \Rightarrow \theta = \cos^{-1}\left(\frac{2}{5}\right)$$

(b)  $\frac{2x+y+3z-2=0}{x}$  and  $\frac{x-2y+5=0}{x}$

$A_1=2, B_1=1, C_1=3$

$A_2=1, B_2=-2, C_2=0$

Cond<sup>n</sup>. for Parallel Planes  $\frac{2}{1} \neq \frac{1}{-2} \neq \frac{3}{0}$  (Not Parallel)

Condition for Perpendicular Planes.  $2(-2) + 0 = 0$

$0=0$  ✓

(Perpendicular planes)



(c)  $2x - 2y + 4z + 5 = 0$  and  $3x - 3y + 6z - 1 = 0$

Condition for Parallel planes

$$\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2} \checkmark$$

$$\Rightarrow \frac{2}{3} = \frac{-2}{-3} = \frac{4}{6} \Rightarrow \frac{2}{3} = \frac{2}{3} = \frac{2}{3}$$

$\therefore$  Parallel planes

(d)  $2x - y + 3z - 1 = 0$  &  $2x - y + 3z + 3 = 0$

Cond<sup>n</sup> for Parallel planes

$$\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2}$$

$$\Rightarrow \frac{2}{2} = \frac{-1}{-1} = \frac{3}{3} \Rightarrow 1 = 1 = 1$$

$\therefore$  Parallel planes

(e)  $4x + 8y + z - 8 = 0$  and  $y + z - 4 = 0$

Cond<sup>n</sup> for Parallel planes

$$\frac{4}{0} \neq \frac{8}{1} \neq \frac{1}{1} \quad \text{Not Parallel}$$

Cond<sup>n</sup> for  $\perp$  Planes

$$A_1A_2 + B_1B_2 + C_1C_2 \neq 0$$

$$\Rightarrow 4 \cdot 0 + 8 \cdot 1 + 1 \cdot 1 \neq 0$$

$$\Rightarrow 0 + 8 + 1 \neq 0$$

(Not  $\perp$  planes)

$$\cos \theta = \frac{0 + 8 + 1}{\sqrt{16 + 64 + 1} \sqrt{0 + 1 + 1}} \Rightarrow \cos \theta = \frac{9}{\sqrt{81} \sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\theta = 45^\circ$$

**Q.14** In the following cases, find the distance of each of the given points from the corresponding given plane:

(a) Point  $(0,0,0)$  ; plane :  $3x - 4y + 12z = 3$

$3x - 4y + 12z - 3 = 0$   $(x, y, z)$   $Ax + By + Cz = d$   
 $\rightarrow Ax + By + Cz - d = 0$

$$d = \left| \frac{0 - 0 + 0 - 3}{\sqrt{9 + 16 + 144}} \right|$$

$$d = \left| \frac{-3}{\sqrt{169}} \right| \Rightarrow d = \frac{3}{13}$$

$$d = \left| \frac{Ax_1 + By_1 + Cz_1 - d}{\sqrt{A^2 + B^2 + C^2}} \right|$$

(b) Point  $(3, -2, 1)$  , plane  $2x - y + 2z + 3 = 0$

$$d = \left| \frac{6 + 2 + 2 + 3}{\sqrt{4 + 1 + 4}} \right| = \frac{13}{\sqrt{9}} = \frac{13}{3}$$

(c) Point  $(2, 3, -5)$  plane :  $x + 2y - 2z = 9$   
 $\Rightarrow x + 2y - 2z - 9 = 0$

$$d = \left| \frac{2 + 6 + 10 - 9}{\sqrt{1 + 4 + 4}} \right| = \frac{9}{\sqrt{9}} = \frac{9}{3} = 3$$

(d) Point  $(-6, 0, 0)$  Plane :  $2x - 3y + 6z - 2 = 0$

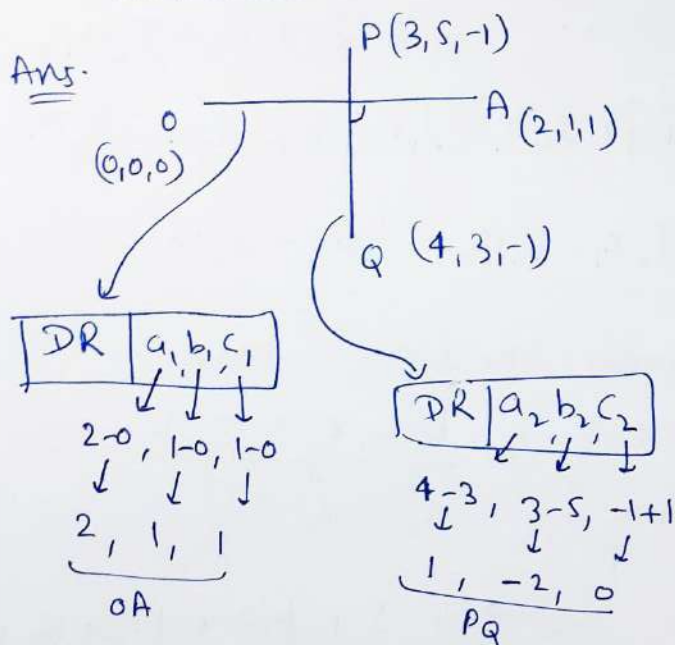
$$d = \left| \frac{-12 + 0 + 0 - 2}{\sqrt{4 + 9 + 36}} \right| = \left| \frac{-14}{\sqrt{49}} \right| = \frac{14}{7} = 2$$



Miscellaneous Exercise on Chapter 11

[Q.1] Show that the line joining the origin to the point  $A(2,1,1)$  is perpendicular to the line determined by the points  $(3,5,-1)$  &  $(4,3,-1)$ .

Ans.



Condition of Perpendicular lines

$$a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$$

$$a_1 a_2 + b_1 b_2 + c_1 c_2$$

$$= 2(1) + 1(-2) + 1(0)$$

$$= 2 - 2 + 0$$

$$= 0$$

Perpendicular lines

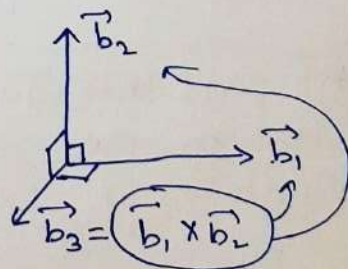
[Q.2] If  $l_1, m_1, n_1$  and  $l_2, m_2, n_2$  are the direction cosines of two mutually perpendicular lines, show that the direction cosines of the line perpendicular to both of these are  $m_1 n_2 - m_2 n_1$ ,  $n_1 l_2 - n_2 l_1$ ,  $l_1 m_2 - l_2 m_1$ .

$$\vec{b}_1 = l_1 \hat{i} + m_1 \hat{j} + n_1 \hat{k}$$

$$(l_1^2 + m_1^2 + n_1^2 = 1) \rightarrow |\vec{b}_1| = 1$$

$$\vec{b}_2 = l_2 \hat{i} + m_2 \hat{j} + n_2 \hat{k}$$

$$(l_2^2 + m_2^2 + n_2^2 = 1) \rightarrow |\vec{b}_2| = 1$$



Now,  $\vec{b}_3 = \vec{b}_1 \times \vec{b}_2$  will be perpendicular to both  $\vec{b}_1$  &  $\vec{b}_2$ .

$$\vec{b}_3 = \vec{b}_1 \times \vec{b}_2 = (l_1 \hat{i} + m_1 \hat{j} + n_1 \hat{k}) \times (l_2 \hat{i} + m_2 \hat{j} + n_2 \hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = \hat{i} (m_1 n_2 - m_2 n_1) - \hat{j} (l_1 n_2 - l_2 n_1) + \hat{k} (l_1 m_2 - l_2 m_1)$$

$$= \hat{i} (m_1 n_2 - m_2 n_1) + \hat{j} (l_2 n_1 - l_1 n_2) + \hat{k} (l_1 m_2 - l_2 m_1)$$

D.C.  $m_1 n_2 - m_2 n_1, l_2 n_1 - l_1 n_2, l_1 m_2 - l_2 m_1$

[Q.3] Find the angle between the lines whose direction ratios are  $a, b, c$  &  $b-c, c-a, a-b$ .

D.R.

$$\cos \theta = \left| \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}} \right| = \left| \frac{a(b-c) + b(c-a) + c(a-b)}{\sqrt{a^2 + b^2 + c^2} \sqrt{(b-c)^2 + (c-a)^2 + (a-b)^2}} \right|$$

$$\cos \theta = \left| \frac{ab - ac + bc - ba + ca - cb}{\sqrt{\quad} \sqrt{\quad}} \right| = \left| \frac{0}{\sqrt{\quad} \sqrt{\quad}} \right|$$

$$\cos \theta = 0$$

$$\theta = 90^\circ$$

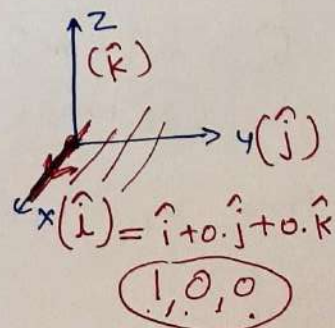
Perpendicular

[Q.4] Find the equation of a line parallel to x-axis & passing through the origin.

Point  $(0, 0, 0)$  Direction:  $(1, 0, 0)$

Eqn. of line  $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$

$$\Rightarrow \left| \frac{x-0}{1} = \frac{y-0}{0} = \frac{z-0}{0} \right| \Rightarrow \frac{x}{1} = \frac{y}{0} = \frac{z}{0}$$





Q.5 If the coordinates of the points A, B, C, D are (1, 2, 3), (4, 5, 7), (-4, 3, -6) and (2, 9, 2) respectively, then find angle b/w the lines AB & CD.

$C(-4, 3, -6)$   $B(4, 5, 7) \Rightarrow DR = 4-1, 5-2, 7-3$   
 $= (3, 3, 4)$   
 $[a_1, b_1, c_1]$   
 $A(1, 2, 3)$   $D(2, 9, 2) \Rightarrow DR = 2-1, 9-2, 2-3$   
 $= (1, 7, -1)$   
 $[a_2, b_2, c_2]$

$$\cos \theta = \left| \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}} \right|$$

$$= \left| \frac{3 \times 1 + 3 \times 7 + 4 \times (-1)}{\sqrt{9+9+16} \sqrt{1+49+1}} \right| = \left| \frac{18+18-4}{\sqrt{34} \sqrt{50}} \right|$$

$$\Rightarrow \cos \theta = \frac{32}{\sqrt{2 \times 17} \sqrt{2 \times 25}} = \frac{32}{\sqrt{2 \times 17 \times 2 \times 25}}$$

$$\Rightarrow \cos \theta = \frac{32}{2 \times 17 \times 5} = 1$$

$$\Rightarrow \cos \theta = 1 \Rightarrow \theta = 0^\circ$$

Q.6 If the lines  $\frac{x-1}{-3} = \frac{y-2}{2K} = \frac{z-3}{2}$  and  $\frac{x-1}{3K} = \frac{y-1}{1} = \frac{z-6}{-5}$  are perpendicular, find K.

Cond<sup>n</sup>. for  $\perp$  lines:

$$a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$$

$$\Rightarrow -9K + 2K - 10 = 0$$

$$\Rightarrow -7K = 10$$

$$\Rightarrow K = \frac{10}{-7}$$

Miscellaneous Exercise on Chapter 11

**Q. 7** Find the vector equation of the line passing through  $(1, 2, 3)$  and perpendicular to the plane  $\vec{r} \cdot (\hat{i} + 2\hat{j} - 5\hat{k}) + 9 = 0$ .

Line:

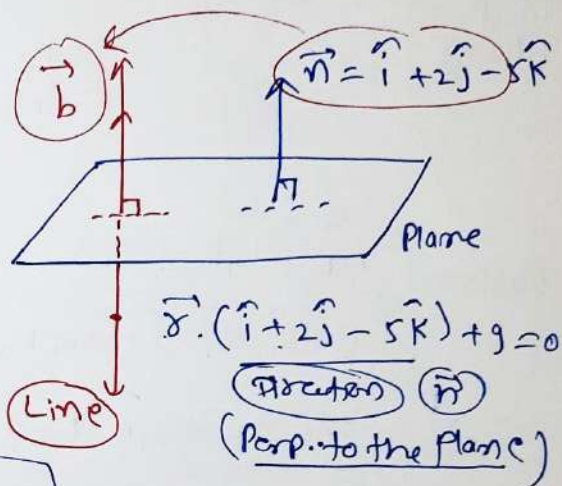
$$\vec{r} = \vec{a} + \lambda \vec{b}$$

Point  $(1, 2, 3)$   
 $(\hat{i} + 2\hat{j} + 3\hat{k})$

Direction  $\vec{b} \parallel \vec{n}$

Line:

$$\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda (\hat{i} + 2\hat{j} - 5\hat{k})$$



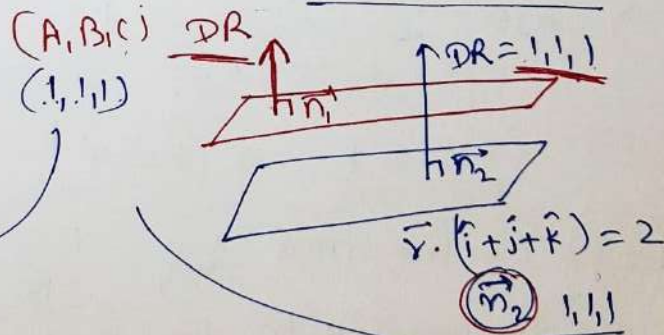
**Q. 8** Find the equation of the plane passing through  $(a, b, c)$  and parallel to the plane  $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 2$ .

Plane passing through  $(x_1, y_1, z_1)$

$$A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$$

DR  $(A, B, C)$   
 $(1, 1, 1)$

$A = B = C = 1$



Required plane

$$1(x - a) + 1(y - b) + 1(z - c) = 0$$

$$\Rightarrow \boxed{x + y + z = a + b + c}$$



Q.9 Find the shortest distance b/w lines

$$\vec{r} = (6\hat{i} + 2\hat{j} + 2\hat{k}) + \lambda(\hat{i} - 2\hat{j} + 2\hat{k}) \quad \& \quad \vec{r} = (-4\hat{i} - \hat{k}) + \mu(3\hat{i} - 2\hat{j} - 2\hat{k})$$

Point ( $\vec{a}_1$ )

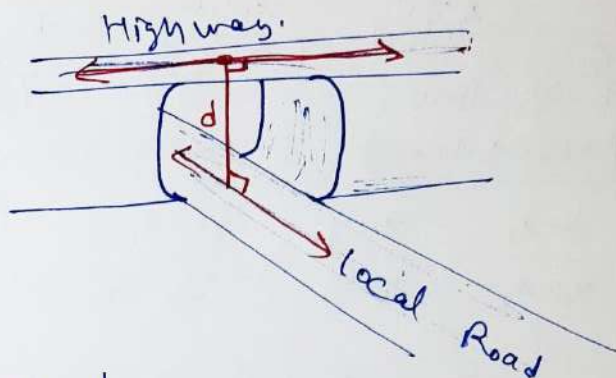
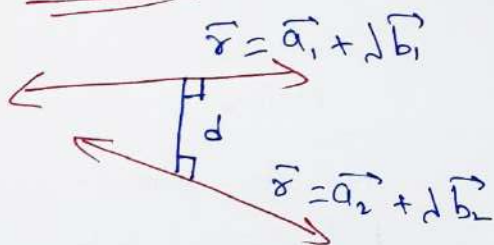
Direction ( $\vec{b}_1$ )

Point ( $\vec{a}_2$ )

Direction ( $\vec{b}_2$ )



Skew Lines



$$d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$$

$$d = \frac{|(-10\hat{i} - 2\hat{j} - 3\hat{k}) \cdot (8\hat{i} + 8\hat{j} + 4\hat{k})|}{12}$$

$$d = \frac{|-80 - 16 - 12|}{12}$$

$$d = \frac{|-108|}{12}$$

$$d = \frac{108}{12}$$

$$d = 9$$

$$\vec{b}_1 \times \vec{b}_2$$

$$= (\hat{i} - 2\hat{j} + 2\hat{k}) \times (3\hat{i} - 2\hat{j} - 2\hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 2 \\ 3 & -2 & -2 \end{vmatrix} \rightarrow R_1$$

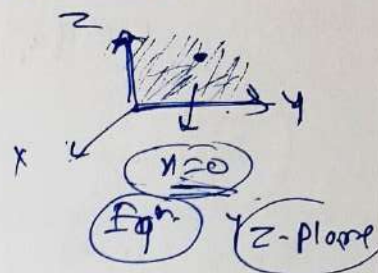
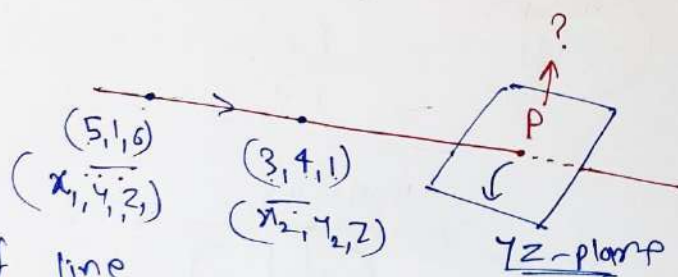
$$= \hat{i}(4 + 4) - \hat{j}(-2 - 6) + \hat{k}(-2 + 6)$$

$$= 8\hat{i} + 8\hat{j} + 4\hat{k} = \vec{b}_1 \times \vec{b}_2$$

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{64 + 64 + 16}$$

$$= \sqrt{144} = 12$$

**Q.10** Find the coordinates of the point where the line through  $(5,1,6)$  and  $(3,4,1)$  crosses the  $yz$ -plane.



Eq<sup>n</sup>. of line

Passing through 2-Points.

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

$$\Rightarrow \frac{x-5}{-2} = \frac{y-1}{3} = \frac{z-6}{-5} = \lambda$$

for 'P'

$$\frac{x-5}{-2} = \lambda \Rightarrow \boxed{x = -2\lambda + 5}$$

$$\frac{y-1}{3} = \lambda \Rightarrow \boxed{y = 3\lambda + 1}$$

$$\frac{z-6}{-5} = \lambda \Rightarrow \boxed{z = -5\lambda + 6}$$

Method-II

For  $yz$ -plane,

Put  $x=0$

$$\frac{0-5}{-2} = \frac{y-1}{3} = \frac{z-6}{-5}$$

Point P  $(-2\lambda + 5, 3\lambda + 1, -5\lambda + 6)$

Satisfies the plane

$(yz\text{-Plane})$

Eq<sup>n</sup>.  $\boxed{x=0}$

$$\Rightarrow -2\lambda + 5 = 0$$

$$\Rightarrow \boxed{\lambda = \frac{5}{2}}$$

Required

Point P  $(-2\lambda + 5, 3\lambda + 1, -5\lambda + 6)$

$$P\left[-2\left(\frac{5}{2}\right) + 5, 3\left(\frac{5}{2}\right) + 1, -5\left(\frac{5}{2}\right) + 6\right]$$

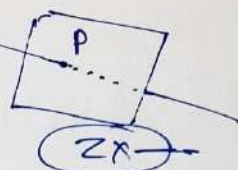
$$P\left(0, \frac{17}{2}, -\frac{13}{2}\right) \checkmark$$



**Q.11** Find the coordinates of the point where the line through  $(5, 1, 6)$  &  $(3, 4, 1)$  crosses the  $Zx$ -plane.

Ans.

line eq<sup>n</sup> → same as  
Previous question.



$y = 0$  ①

By solving eq<sup>n</sup> ① & ②

$$\frac{x-5}{-2} = \frac{y-1}{3} = \frac{z-6}{-5} \quad \text{--- ②}$$

$$\frac{x-5}{-2} = \frac{0-1}{3} = \frac{z-6}{-5}$$

$$\frac{x-5}{-2} = \frac{-1}{3} \Rightarrow x-5 = \frac{2}{3} \Rightarrow x = 5 + \frac{2}{3} \Rightarrow \boxed{x = \frac{17}{3}}$$

$$\frac{-1}{3} = \frac{z-6}{-5} \Rightarrow \frac{5}{3} + 6 = z \Rightarrow \boxed{\frac{23}{3} = z}$$

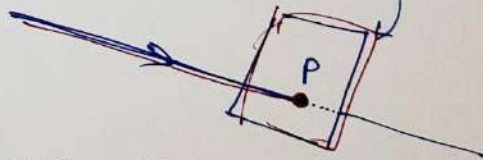
Point of intersection  $P\left(\frac{17}{3}, 0, \frac{23}{3}\right)$

**Q.12** Find the coordinates of the point where the line through  $(3, -4, -5)$  and  $(2, -3, 1)$  crosses the plane

$$2x + y + z = 7.$$

Line:  $\frac{x-3}{-1} = \frac{y+4}{1} = \frac{z+5}{6} = \lambda$

$$\begin{aligned} \Rightarrow \frac{x-3}{-1} &= \lambda & \frac{y+4}{1} &= \lambda & z+5 &= 6\lambda \\ \Rightarrow \underline{x} &= -\lambda+3 & \Rightarrow \underline{y} &= \lambda-4 & \underline{z} &= 6\lambda-5 \end{aligned}$$



$P(-\lambda+3, \lambda-4, 6\lambda-5) \Rightarrow$  Also satisfies the plane

Point of intersection  $P(-\lambda+3, \lambda-4, 6\lambda-5)$

Satisfies the plane  $\underline{2x+y+z=7}$ .

$$\Rightarrow 2(-\lambda+3) + (\lambda-4) + (6\lambda-5) = 7$$

$$\Rightarrow \underline{-2\lambda+6} + \underline{7\lambda-9} = 7$$

$$\Rightarrow 5\lambda = 10$$

$$\boxed{\lambda=2}$$

$$P(-2+3, 2-4, 12-5)$$

$$\underline{\underline{P(1, -2, 7)}}$$



Miscellaneous Exercise on Chapter (11)

**Q.13** Find the equation of the plane passing through the point  $(-1, 3, 2)$  and perpendicular to each of the planes  $x+2y+3z=5$  and  $3x+3y+z=0$

Required plane, (P)  $A(x-x_1)+B(y-y_1)+C(z-z_1)=0$   
 $(x_1, y_1, z_1) \equiv (-1, 3, 2)$

$P: \boxed{A(x+1)+B(y-3)+C(z-2)=0}$   $DR_p = A, B, C$

$P_1: \boxed{x+2y+3z=5} \rightarrow DR_{P_1} = 1, 2, 3$   $(\vec{n}_1 = \hat{i} + 2\hat{j} + 3\hat{k})$

$P_2: \boxed{3x+3y+z=0} \rightarrow DR_{P_2} = 3, 3, 1$   $(\vec{n}_2 = 3\hat{i} + 3\hat{j} + \hat{k})$

ATQ Condition for Perpendicular Planes.

$\boxed{A_1A_2+B_1B_2+C_1C_2=0}$

$\boxed{P \perp P_1}$

$Ax_1 + Bx_2 + Cx_3 = 0$

$\Rightarrow A + 2B + 3C = 0$

$\Rightarrow \boxed{A = -2B - 3C}$

$\Rightarrow A = -2\left(-\frac{8C}{3}\right) - 3C$

$\Rightarrow A = \frac{16C}{3} - 3C$

$\Rightarrow \boxed{A = \frac{7C}{3}}$  ✓

$P \perp P_2$

$3A + 3B + C = 0$

$\Rightarrow 3(-2B - 3C) + 3B + C = 0$

$\Rightarrow -6B - 9C + 3B + C = 0$

$\Rightarrow -3B - 8C = 0$

$\Rightarrow -3B = 8C$

$\Rightarrow \boxed{B = -\frac{8C}{3}}$  ✓

Required plane  $A(x+1) + B(y-3) + C(z-2) = 0$

$$A = \frac{7C}{3}, \quad B = \frac{-8C}{3}$$

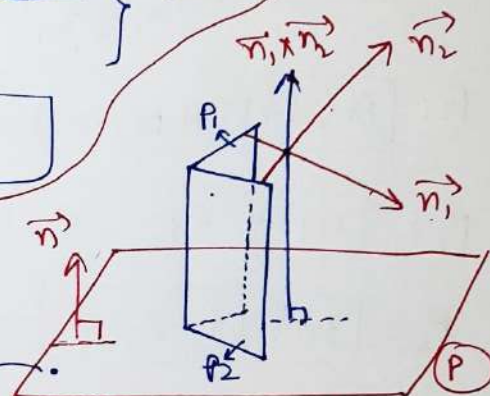
$$\Rightarrow \frac{7C}{3}(x+1) - \frac{8C}{3}(y-3) + C(z-2) = 0$$

$$\Rightarrow \textcircled{C} \left\{ \frac{7x+7-8y+24+3z-6}{3} \right\} = 0$$

$$\Rightarrow 7x - 8y + 3z + 25 = 0$$

II - method  
(Hint only)  
vector method

$(-1, 3, 2)$



Required plane

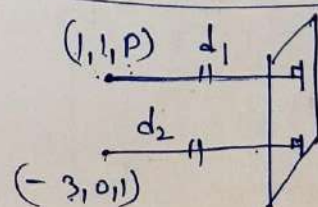
$$\vec{r} \cdot \vec{n} = d$$

normal =  $\vec{n}_1 \times \vec{n}_2$

Q.14 If the points  $(1, 1, P)$  and  $(-3, 0, 1)$  be equidistant from the ~~point~~ plane  $\vec{r} \cdot (3\hat{i} + 4\hat{j} - 12\hat{k}) + 13 = 0$ , then find the value of P.

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$3x + 4y - 12z + 13 = 0$$



$$d = \frac{|Ax_1 + By_1 + Cz_1 - d|}{\sqrt{A^2 + B^2 + C^2}}$$



ATQ.  $d_1 = d_2$

$$\Rightarrow \left| \frac{3+4-12P+13}{\sqrt{9+16+144}} \right| = \left| \frac{-9+0-12+13}{\sqrt{9+16+144}} \right|$$

$$\Rightarrow \underline{\underline{|20-12P| = |-8| = 8}}$$

$$\Rightarrow \boxed{20-12P = \pm 8}$$

$$20-12P = 8$$

$$\Rightarrow 12 = 12P$$

$$\Rightarrow \boxed{P=1} \checkmark$$

$$20-12P = -8$$

$$\Rightarrow 28 = 12P$$

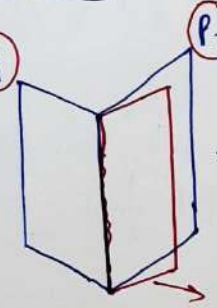
$$\Rightarrow \boxed{\frac{7}{3} = P} \checkmark$$

**Q.15** Find the equation of the plane passing through the line of intersection of the planes

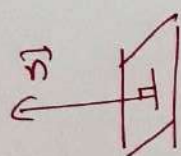
$\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 1$  and  $\vec{r} \cdot (2\hat{i} + 3\hat{j} - \hat{k}) + 4 = 0$  and parallel to x-axis.

$\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 1$   $P_1$   
 $\Rightarrow x+y+z=1$   
 $\Rightarrow \boxed{x+y+z-1=0}$   $P_1$

$\vec{r} \cdot (2\hat{i} + 3\hat{j} - \hat{k}) + 4 = 0$   $P_2$   
 $\Rightarrow \boxed{2x+3y-z+4=0}$



New plane  $P_1 + \lambda P_2 = 0$

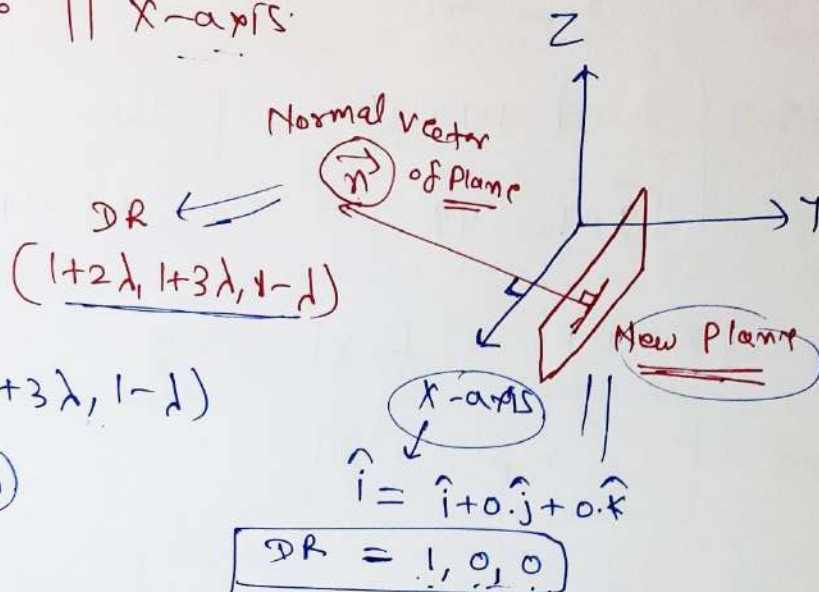


$$\Rightarrow (x+y+z-1) + \lambda(2x+3y-z+4) = 0$$

$$\Rightarrow \boxed{x(1+2\lambda) + y(1+3\lambda) + z(1-\lambda) + (4\lambda-1) = 0}$$

DR  $\rightarrow \underline{1+2\lambda}, \underline{1+3\lambda}, \underline{1-\lambda}$

New plane  $\parallel$  X-axis.



$$\vec{n} \rightarrow (1+2\lambda, 1+3\lambda, 1-\lambda)$$

$$\text{X-axis} \rightarrow (1, 0, 0)$$

By Diagram,

$\vec{n} \perp$  X-axis.

Condition  $a_1a_2 + b_1b_2 + c_1c_2 = 0$

$$\Rightarrow (1+2\lambda) \cdot 1 + (1+3\lambda) \cdot 0 + (1-\lambda) \cdot 0 = 0$$

$$\Rightarrow 1+2\lambda = 0$$

$$\Rightarrow \lambda = -\frac{1}{2}$$

New plane,  $(1+2\lambda)x + (1+3\lambda)y + (1-\lambda)z + (4\lambda-1) = 0$

Put  $\left(\lambda = -\frac{1}{2}\right)$

$$\Rightarrow 0 \cdot x + \left(1 - \frac{3}{2}\right)y + \left(1 + \frac{1}{2}\right)z + \left(\frac{-4}{2} - 1\right) = 0$$

$$\Rightarrow \left(-\frac{1}{2}y\right) + \left(\frac{3}{2}z\right) = 3$$

$$\Rightarrow \frac{-y + 3z}{2} = 3$$

$$\Rightarrow -y + 3z = 6 \Rightarrow 0 = y - 3z + 6$$

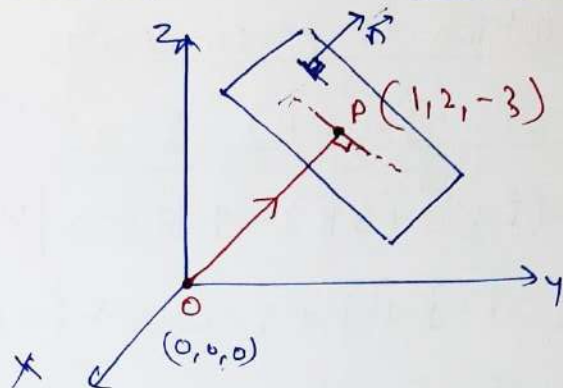


[Q.16] If 'O' be the origin and the coordinates of 'P' be  $P(1, 2, -3)$ , then find the equation of the plane passing through P and perpendicular to OP.

$$O(0, 0, 0)$$

$$P(1, 2, -3)$$

$$\text{DR}_{OP} = 1, 2, -3$$



Any plane passing through the point  $(x_1, y_1, z_1)$

$$Eq^n \Rightarrow A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$$

$$P(1, 2, -3)$$

Plane:  $A(x - 1) + B(y - 2) + C(z + 3) = 0$

$$\text{DR} = \underline{A, B, C} \leftarrow \text{DR of normal vector of Plane}$$

By Diagram,  $\vec{OP} \parallel \vec{n}$

$$\text{DR} \rightarrow \begin{array}{|c|c|} \hline 1, 2, -3 & A, B, C \\ \hline \end{array}$$

$$\underline{A=1}, \underline{B=2}, \underline{C=-3}$$

Required Plane:  $1(x - 1) + 2(y - 2) + (-3)(z + 3) = 0$

$$\Rightarrow x - 1 + 2y - 4 - 3z - 9 = 0$$

$$\Rightarrow \boxed{x + 2y - 3z = 14}$$

[Q.17] Find the equation of the plane which contains the line of intersection of the planes  $\vec{r} \cdot (\hat{i} + 2\hat{j} + 3\hat{k}) - 4 = 0$ ,  $\vec{r} \cdot (2\hat{i} + \hat{j} - \hat{k}) + 5 = 0$  and which is perpendicular to the plane

$$\vec{r} \cdot (5\hat{i} + 3\hat{j} - 6\hat{k}) + 8 = 0 \rightarrow P_3$$

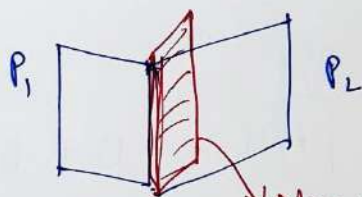
$$P_1: \vec{r} \cdot (\hat{i} + 2\hat{j} + 3\hat{k}) - 4 = 0 \Rightarrow \boxed{x + 2y + 3z - 4 = 0} P_1$$

$$P_2: \vec{r} \cdot (2\hat{i} + \hat{j} - \hat{k}) + 5 = 0 \Rightarrow \boxed{2x + y - z + 5 = 0} P_2$$

$$P_3: \vec{r} \cdot (5\hat{i} + 3\hat{j} - 6\hat{k}) + 8 = 0 \Rightarrow \boxed{5x + 3y - 6z + 8 = 0} P_3$$

New plane  $\rightarrow$  passing through the line of intersection of  $P_1$  &  $P_2$

$$\boxed{P_1 + \lambda P_2 = 0}$$



$$\Rightarrow (x + 2y + 3z - 4) + \lambda(2x + y - z + 5) = 0$$

$$\Rightarrow \boxed{x(1 + 2\lambda) + y(2 + \lambda) + z(3 - \lambda) + (5\lambda - 4) = 0} \text{ New plane}$$

$$DR(1 + 2\lambda, 2 + \lambda, 3 - \lambda)$$

$\downarrow$  This New plane is perpendicular to the plane  $P_3: \boxed{5x + 3y - 6z + 8 = 0}$

Condition of Perpendicular planes.

$$DR: \underline{5, 3, -6}$$

$$A_1A_2 + B_1B_2 + C_1C_2 = 0$$

$$\Rightarrow (1 + 2\lambda) \cdot 5 + (2 + \lambda) \cdot 3 + (3 - \lambda) \cdot (-6) = 0$$

$$\Rightarrow \underline{5} + \underline{10\lambda} + \underline{6} + \underline{3\lambda} - \underline{18} + \underline{6\lambda} = 0$$

$$\Rightarrow 19\lambda = 7 \Rightarrow \boxed{\lambda = \frac{7}{19}}$$



New plane :  $x(1+2\lambda) + y(2+\lambda) + z(3-\lambda) + (5\lambda-4) = 0$

Put  $(\lambda = \frac{7}{19}) \uparrow$

$$\Rightarrow x\left(1 + \frac{14}{19}\right) + y\left(2 + \frac{7}{19}\right) + z\left(3 - \frac{7}{19}\right) + \left(\frac{35}{19} - 4\right) = 0$$

$$\Rightarrow x\left(\frac{33}{19}\right) + y\left(\frac{45}{19}\right) + z\left(\frac{50}{19}\right) + \left(-\frac{41}{19}\right) = 0$$

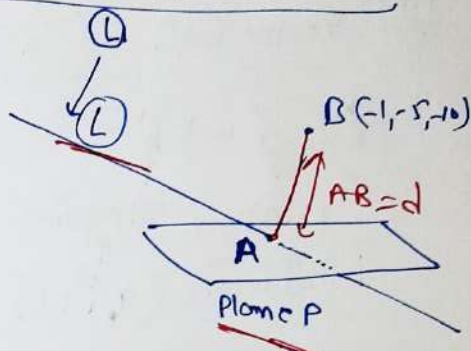
$$\Rightarrow \boxed{33x + 45y + 50z - 41 = 0}$$

Miscellaneous Exercise on Chapter 11

Q.18 Find the distance of the point  $B(-1, -5, -10)$  from the point of intersection of the line  $\vec{r} = (2\hat{i} - \hat{j} + 2\hat{k}) + \lambda(3\hat{i} + 4\hat{j} + 2\hat{k})$  and the plane  $\vec{r} \cdot (\hat{i} - \hat{j} + \hat{k}) = 5$ .

Plane P

A  $\rightarrow$  Point of intersection of line L & plane P



We solve the equations of line L & plane P for the point of intersection.

Line L:  $\vec{r} = (2\hat{i} - \hat{j} + 2\hat{k}) + \lambda(3\hat{i} + 4\hat{j} + 2\hat{k})$  — (1)

Plane P:  $\vec{r} \cdot (\hat{i} - \hat{j} + \hat{k}) = 5$  — (2)

Substitution method.

$$\Rightarrow [(2\hat{i} - \hat{j} + 2\hat{k}) + \lambda(3\hat{i} + 4\hat{j} + 2\hat{k})] \cdot (\hat{i} - \hat{j} + \hat{k}) = 5$$

(Dot Product)

$$\Rightarrow 2 + 1 + 2 + \lambda(3 - 4 + 2) = 5$$

$$\Rightarrow 5 + \lambda(1) = 5$$

$$\Rightarrow \boxed{\lambda = 0}$$

By eqn (1):

$$\vec{r} = 2\hat{i} - \hat{j} + 2\hat{k}$$

Point of intersection (A)

$$A(2, -1, 2)$$

$$AB = \sqrt{(3)^2 + 4^2 + 12^2} = \sqrt{9 + 16 + 144} = \sqrt{169} = 13$$

B(-1, -5, -10) — Given





Q.19 Find the vector equation of the line passing through  $(1, 2, 3)$  and parallel to the planes  $\vec{r} \cdot (\hat{i} - \hat{j} + 2\hat{k}) = 5$  and  $\vec{r} \cdot (3\hat{i} + \hat{j} + \hat{k}) = 6$ .

Ans. Vector Eq<sup>n</sup>. of a line.

$$\vec{r} = \underset{\substack{\uparrow \\ \text{Point}}} {\vec{a}} + \lambda \underset{\substack{\uparrow \\ \text{Direction}}} {\vec{b}}$$

$$\hat{i} + 2\hat{j} + 3\hat{k} \quad (a\hat{i} + b\hat{j} + c\hat{k})$$

$$\Rightarrow \boxed{\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda (a\hat{i} + b\hat{j} + c\hat{k})} \quad \text{--- (B)}$$

Our required line is parallel to the plane  $(P_1)$

$$\vec{r} \cdot (\hat{i} - \hat{j} + 2\hat{k}) = 5$$

$$\Rightarrow \boxed{\vec{b} \cdot \vec{n}_1 = 0}$$

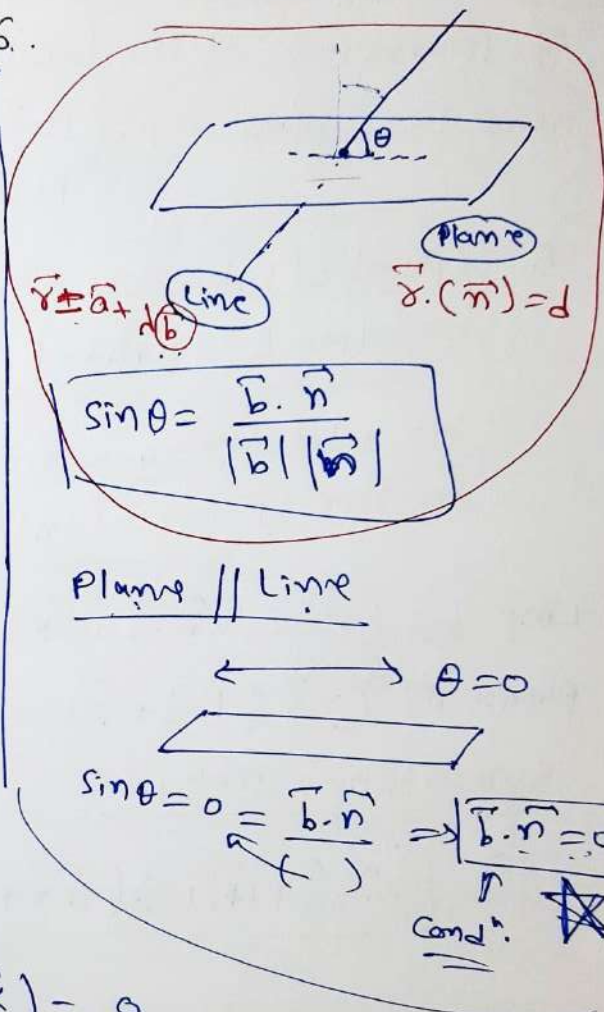
$$\Rightarrow (a\hat{i} + b\hat{j} + c\hat{k}) \cdot (\hat{i} - \hat{j} + 2\hat{k}) = 0$$

$$\Rightarrow \boxed{a - b + 2c = 0} \quad \text{--- (1)}$$

$$\text{Line } L \parallel \vec{r} \cdot (3\hat{i} + \hat{j} + \hat{k}) = 6$$

$$\Rightarrow (a\hat{i} + b\hat{j} + c\hat{k}) \cdot (3\hat{i} + \hat{j} + \hat{k}) = 0$$

$$\Rightarrow \boxed{3a + b + c = 0} \quad \text{--- (2)}$$



By eq<sup>n</sup> (1) & (2)

Elimination.

$$a - b + 2c = 0$$

$$3a + b + c = 0$$

$$+$$

$$4a + 3c = 0$$

$$\boxed{c = -\frac{4a}{3}}$$

by putting the value of  $c = -\frac{4a}{3}$  in eqn (1)

$$\Rightarrow a + 2c = b$$

$$\Rightarrow a + 2\left(-\frac{4a}{3}\right) = b$$

$$\Rightarrow \frac{3a - 8a}{3} = b \Rightarrow \boxed{b = -\frac{5a}{3}}$$

$$\boxed{a - b + 2c = 0}$$

Required Pq<sup>n</sup>. of line  $\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(a\hat{i} + b\hat{j} + c\hat{k})$

$$\Rightarrow \vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda\left(\frac{a}{1}\hat{i} - \frac{5a}{3}\hat{j} - \frac{4a}{3}\hat{k}\right)$$

$$\Rightarrow \vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda\left(\frac{3a\hat{i} - 5a\hat{j} - 4a\hat{k}}{3}\right)$$

$$\Rightarrow \vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \underbrace{\lambda\left(-\frac{a}{3}\right)}_{\lambda_1} (-3\hat{i} + 5\hat{j} + 4\hat{k})$$

$$\Rightarrow \boxed{\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda_1 (-3\hat{i} + 5\hat{j} + 4\hat{k})}$$

[Q.20] Find the vector equation of the line passing through the point  $(1, 2, -4)$  and perpendicular

to the lines  $\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$  and  $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$

Vector Eq<sup>n</sup>. of line (L)

$$\vec{r} = \vec{a} + \lambda \vec{b}$$

Point

Direction

$$(\hat{i} + \hat{j} - 4\hat{k})$$

$$(a\hat{i} + b\hat{j} + c\hat{k})$$

Direction

$$\vec{b}_1 = 3\hat{i} - 16\hat{j} + 7\hat{k}$$

Concept

$$\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$$

$$\vec{r} = \vec{a}_2 + \mu \vec{b}_2$$

$\perp$

Direction

$$\vec{b}_2 = 3\hat{i} + 8\hat{j} - 5\hat{k}$$

$$\vec{b}_1 \cdot \vec{b}_2 = 0$$



ATQ Line  $L \perp L_1$   
 $(\vec{b} = a\hat{i} + b\hat{j} + c\hat{k})$   $(\vec{b}_1 = 3\hat{i} - 16\hat{j} + \underline{7\hat{k}})$

$$\Rightarrow \vec{b} \cdot \vec{b}_1 = 0$$

$$\Rightarrow \boxed{3a - 16b + 7c = 0} \quad \text{--- (1)}$$

ATQ Line  $L \perp L_2$   
 $(\vec{b} = a\hat{i} + b\hat{j} + c\hat{k})$   $(\vec{b}_2 = 3\hat{i} + 8\hat{j} - 5\hat{k})$

$$\Rightarrow \vec{b} \cdot \vec{b}_2 = 0$$

$$\Rightarrow \boxed{3a + 8b - 5c = 0} \quad \text{--- (2)}$$

By eqn. (1) - eqn. (2)

$$\begin{array}{r} 3a - 16b + 7c = 0 \\ 3a + 8b - 5c = 0 \\ \hline -24b + 12c = 0 \end{array}$$

$$\Rightarrow 12c = \cancel{24} \frac{b}{2}$$

$$\boxed{c = 2b}$$

By eqn. (2)

$$\Rightarrow 3a + 8b - 5(2b) = 0$$

$$\Rightarrow 3a + 8b - 10b = 0$$

$$\Rightarrow 3a = 2b \Rightarrow \boxed{a = \frac{2b}{3}}$$

Required line  $\boxed{\vec{r} = \vec{a} + \lambda \vec{b}}$

$$\Rightarrow \vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda (\underline{a\hat{i} + b\hat{j} + c\hat{k}})$$

$$\Rightarrow \vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda (\underline{\frac{2b}{3}\hat{i} + b\hat{j} + 2b\hat{k}})$$

$$\Rightarrow \vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \underline{b\lambda} (\underline{\frac{2\hat{i} + 3\hat{j} + 6\hat{k}}{2}})$$

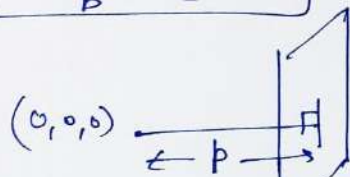
$$\Rightarrow \boxed{\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda_1 (2\hat{i} + 3\hat{j} + 6\hat{k})}$$

New  $\lambda_1$

Q.21) Prove that if a plane has intercepts  $a, b, c$  and is at a distance ' $p$ ' units from the origin, then  $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{p^2}$ .

Intercept form of the plane.

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

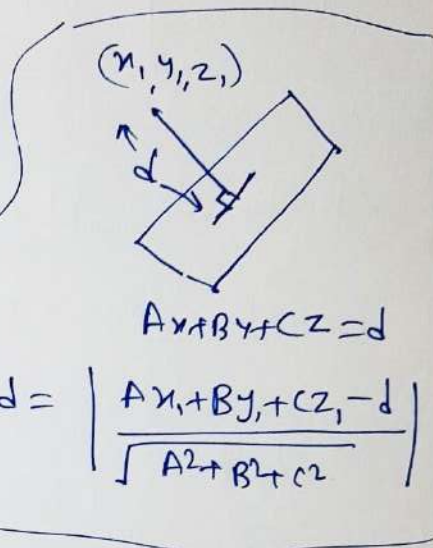


$$P = \left| \frac{0+0+0-1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}} \right|$$

$$\Rightarrow P = \frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}}$$

$$\Rightarrow \frac{1}{p} = \sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}$$

$$\Rightarrow \frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$$



$$Ax + By + Cz = d$$

$$d = \left| \frac{Ax_1 + By_1 + Cz_1 - d}{\sqrt{A^2 + B^2 + C^2}} \right|$$

Q.22) Distance b/w the two planes:  $2x + 3y + 4z = 4$  (P<sub>1</sub>) &  $4x + 6y + 8z = 12$  (P<sub>2</sub>) is —

(A) 2 units (B) 4 (C) 8 (D)  $\frac{2}{\sqrt{29}}$

$$P_1: 2x + 3y + 4z - 4 = 0$$

$$P_2: 4x + 6y + 8z - 12 = 0$$

A B C

I-method.

$$d = \left| \frac{d_1 - d_2}{\sqrt{A^2 + B^2 + C^2}} \right|$$

$$d = \left| \frac{(-4) - (-6)}{\sqrt{2^2 + 3^2 + 4^2}} \right| = \frac{2}{\sqrt{29}}$$

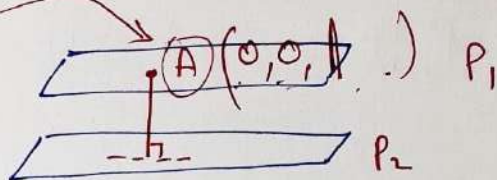
$$(4, 9, 16)$$



II - method

$$P_1: 2x + 3y + 4z - 4 = 0$$

$$P_2: 2x + 3y + 4z - 6 = 0$$



Point A on 'P<sub>1</sub>'  
 $\downarrow$   
 $(0, 0, 1)$

let Point  $(0, 0, z)$   
 on 'P<sub>1</sub>'  
 $\Rightarrow 0 + 0 + 4z - 4 = 0$   
 $\Rightarrow \boxed{z = 1}$

Distance b/w P<sub>1</sub> & P<sub>2</sub>

= Distance b/w A & P<sub>2</sub>

(Point) (Plane)  
 $(0, 0, 1)$

$$\boxed{2x + 3y + 4z - 6 = 0}$$

$$= \left| \frac{0 + 0 + 4 - 6}{\sqrt{2^2 + 3^2 + 4^2}} \right| = \frac{2}{\sqrt{29}}$$

- Q.23 The planes:  $\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2}$   $2x - y + 4z = 5$  and  $5x - 2.5y + 10z = 6$  are (A) perpendicular (B) Parallel (C) intersect y-axis (D) passes through  $(0, 0, \frac{5}{4})$

Condition of Parallel planes

$$\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2}$$

$$\Rightarrow \frac{2}{5} = \frac{+10}{+2.5} = \frac{4}{10}$$

$$\Rightarrow \frac{2}{5} = \frac{+10}{2.5} = \frac{4}{10}$$

$$\frac{2}{5} = \frac{2}{5} = \frac{2}{5}$$